Final Year Project Catalogue 2025-2026

Dr Matthew Pike

School of Computer Science

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Contents

Project Selection

Welcome to the Final Year Projects (FYP) module in the Computer Science Project Catalogue. This resource offers a broad selection of project ideas proposed by our FYP supervisors. These ideas are intended as general guidance, rather than fixed project specifications.

You are encouraged to review the catalogue carefully and identify supervisors whose projects interest you. The aim is to match you with a suitable project supervisor.

Tips for Selecting a Project

- Review and critically analyse all projects listed in the catalogue.

- Highlight those that interest you and carry out some background reading, making brief notes.

- Contact the supervisors of your chosen projects by email. Once a conversation has begun, decide whether it's best to meet in person or online (e.g., via Microsoft Teams).

- You are welcome to propose your own project ideas. However, it's important to discuss them with relevant staff as early as possible.

Project Allocation Process

- **30 April 2025**: Project Catalogue is released
- **30 April - 02 June 2025**: Contact Window Period
  - Students should contact supervisors expressing interest in a particular FYP project, first via email; then with a face-to-face (or video/voice call) meeting.
  - If you are currently a remote learning student, you should arrange to discuss with prospective supervisors via Microsoft Teams.
  - During the *contact window* period, no supervisor can agree to (definitely) supervise a student. This is in order to preserve fairness - allowing ample time for all students to contact potential supervisors.
- **02 June 2025 - Beginning of Autumn Semester**: Project Supervision Confirmation Period
  - Supervisors can formally confirm supervision arrangements by:
    * Completing a digital form and submitting it to the module convener
    * The module convener then emails the student and supervisor to confirm the supervision details

  - Note: Project allocation is only confirmed after this email is sent.
- **08 Oct 2025**: Project allocation ends.
  - The deadline has been updated to account for university holidays.

Important: Each supervisor is (initially) capped at supervising 5 students. This limit may change in accordance to the number of students deciding to take the FYP module.

Students are expected to independently identify and secure the agreement of a supervisor for their project. If students have not confirmed their supervision arrangements before the start of the Autumn semester, they will be encouraged to enroll in taught modules instead of pursuing a FYP.

Course Accreditation: British Computer Society

Some programmes have, or have been recommended for, British Computer Society (BCS) accreditation for Chartered IT Professional (CITP). It is expected that within an undergraduate programme, you will undertake a major computing project, normally in the final year and normally as an individual activity, to meet the BCS accreditation requirements, giving you the opportunity to demonstrate:

- ability to apply practical and analytical skills present in the programme as a whole
- innovation and/or creativity
- synthesis of information, ideas, and practices to provide a quality solution together with an evaluation of that solution
- that the project meets a real need in a wider context
- ability to self-manage a significant piece of work
- critical self-evaluation of the process

Projects must also be passed with a minimum mark of 40%. For further information, please refer to the British Computer Society Guidelines. The BCS website also provides an overview of the benefits of accreditation.

Also note that you must undertake a project in Computer Science that is relevant to your programme of study; for example, projects undertaken by Artificial Intelligence (AI) students must have a strong AI focus. In the catalogue, you will see that projects have been designated as either being suitable or unsuitable for CSAI students.

Ministry of Education: Degree Certification (学位证)

Starting from the academic year 2021-2022, it is mandatory for all students who wish to receive a degree certificate issued by the Ministry of Education (学位证) in China to complete the dissertation module. Please be advised that this requirement does not affect the terms and conditions under which your University of Nottingham degree certificate is issued since it is under the administration of the UK-based campus.

Anthony Bellotti 's Project Ideas

General Information

I am an Associate Professor in the School of Computer Science, working in the AI and Optimization Research Group. I have an interest in machine learning applied to credit risk and consumer finance, as well as conformal prediction for reliable machine learning.

If you are interested in one of the projects, please send me an email to make an appointment to discuss.

- Email: anthony-graham.bellotti@nottingham.edu.cn
- Office: PMB419

Projects

[AB-1] Algorithm for Density Estimation by Classification

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Density estimation remains a difficult computational task. The standard existing technique using kernel density estimation works well in one-dimension but does not scale well to high-dimensional data which is typical of modern real-world data sets. This project will explore transforming the density estimation problem into a classification problem using probability theory and Bayes' rule. This then allows the use of any state-of-the-art classifiers (such as neural networks) to perform density estimation. This project will develop this algorithm and evaluate the method on simulated and real-world data, such as fraud detection, comparing against traditional methods and some other state-of-the-art solutions using deep learning. All programming will be in Python. This project is suitable for a student who enjoys mathematics and theory as well as programming.

[AB-2] Automated valuation models: Explainable price forecasts

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Automated valuation models (AVM) are used by financial institutions and property listing sites to help value properties for sale. Traditionally valuations are undertaken by surveyors, human experts. The use of automation using machine learning yields transparent, objective and accurate forecasts.

Currently, AVMs are typically black box models. This is a problem for banks, real-estate agents and individuals who want an explanation of the price. In this project, students will develop techniques that provide useful textual explanations. The methodology must go beyond standard Explainable AI techniques like LIME or SHAP analysis, to provide application-specific results for lay-users. A particular requirement is that the end-user can see how changes to the property will affect the price (e.g. adding an extension). Approaches may be based on rule-based analysis of the black box model and use of large language models to supplement model output. Several property price data sets are available and the student will work in Python.

[AB-3] Explaining interaction effects in credit risk models

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Current research shows that machine learning models perform very well for credit risk modelling. However, banks and other financial institutions are uncomfortable deploying them since the black box models do not offer explanations of decisions. This is required for risk management and, often, is required by legislation. Existing Explainable AI techniques like LIME and SHAP are useful, but they do not give the full picture. In particular, it is believed that success of machine learning in credit risk is because it is able to model interactions between variables and discover heterogeneous segments in data. In this project, the student will explore novel approaches to discover and disclose these patterns, whilst also seeking shifts in the data which may supersede the model. Several credit risk data sets are available and the student will work in Python.

[AB-4] Visualization of Markov Transition Models for Credit Risk

- Suitable for CSAI students: No
- Project type: Software Engineering

Markov Transition Models are popular for modelling credit delinquency and estimating default rates on credit portfolios (roll-rate models). By supplementing them with neural networks they may be enriched with further credit risk features. In this project, you will build a Markov model with neural networks, and develop software to enable a bank analyst to use and interrogate the model to determine its historical performance, achieve dynamic credit risk assessments at individual level, as well as default rate forecasts for portfolio or sub-groups of loans. The software should also allow for stress tests using the model, taking account different economic scenarios. Several credit risk data sets are available and the student will work in Python.

[AB-SP] Student Proposed Project

Very happy to consider student proposed project. Please initially approach me by email.

Alejandro Guerra Manzanares 's Project Ideas

General Information

My research interests lie at the intersection of machine learning, cybersecurity, and healthcare. My primary focus is on applying machine learning to address cybersecurity challenges such as malware detection and intrusion detection. In addition, I am working on privacy-preserving machine learning for healthcare, ensuring that machine learning-based healthcare applications comply with privacy constraints, leveraging techniques such as federated learning. The security of machine learning models, particularly in the context of adversarial machine learning, also falls within the scope of my research. Finally, any topics related to digital forensics, whether or not they involve the use of AI techniques, are of interest to me.

If you'd like to discuss a project or propose a new one, please feel free to reach out to me!

- Email: alejandro.guerra@nottingham.edu.cn
- Office: PMB435

Projects

[AM-1] Development of a Medium-Interaction IoT Honeypot for Cyber Threat Detection

- Suitable for CSAI students: Yes
- Project type: Software Engineering

This project aims to design and implement a medium-interaction Internet of Things (IoT) honeypot to detect and analyse cyber threats targeting IoT devices. The honeypot simulates the behaviour of common IoT devices (e.g., smart cameras, thermostats, or routers) to attract potential attackers while maintaining a controlled environment that limits full system compromise. Unlike low-interaction honeypots that emulate only basic device responses, the medium-interaction honeypot provides richer interaction and mimics real-world device functionality, enabling deeper insights into adversarial techniques, such as credential brute-forcing, firmware exploitation, and malware deployment.

[AM-2] Development of an Android Malware Detection Portal (website)

- Suitable for CSAI students: Yes
- Project type: Software Engineering

This project involves building an Android malware detection portal (website) where users can upload APK files to determine whether the application is malicious or benign. The system leverages advanced machine learning models trained on behavioural, static, and dynamic features extracted from Android applications, such as permissions, API calls, and network activity. Users interact with a user-friendly web interface to upload APK files. The backend processes the files for feature extraction and uses the trained model to classify the app as malware or not, providing users with a detailed report on potential threats and risk levels. The portal ensures minimal latency for real-time results while maintaining user privacy and data security.

[AM-3] Implementation of a Vertical Federated Learning System for Privacy-Preserving Collaborative Data Analysis

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on developing a vertical federated learning (VFL) framework to enable privacy-preserving collaboration between multiple institutions or organizations that hold complementary data features about the same set of users. For example, a healthcare provider could collaborate with a financial institution to jointly train a predictive model where healthcare records and financial data overlap for the same individuals, enabling advanced analytics or decision-making. Unlike horizontal federated learning, which shares model updates based on identical feature spaces, vertical federated learning securely combines complementary feature spaces while keeping sensitive user data private. This project will implement secure computation protocols such as encryption, secret sharing, or privacy-preserving mechanisms to ensure data confidentiality during the training process. The outcome will be a globally trained model that benefits from the shared complementary data features without exposing sensitive information from participating organizations.

[AM-4] Development of a Visually Explainable Malware Detection System

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on creating a visually explainable malware detection system that uses machine learning to classify applications or files as benign or malicious while providing intelligible visual insights into the decision-making process. The system integrates explainability frameworks, such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations), to generate visualizations that highlight key features or behaviors contributing to the classification, such as suspicious API calls, permissions usage, or code patterns. The system's user-friendly interface presents analyses through heatmaps, graphs, or feature importance plots, allowing security analysts or non-expert users to better understand why a file or application is flagged as malware. This enhances trust in machine learning models, facilitates debugging, and supports decision-making in cybersecurity processes.

[AM-5] Leveraging Large Language Models (LLMs) for Cyber Threat Intelligence and Defense

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project explores the application of large language models (LLMs) to enhance cybersecurity tasks like threat detection, incident response, and vulnerability assessment. The LLM-based system utilizes natural language understanding to analyze unstructured threat intelligence reports, identify emerging attack patterns, and extract actionable insights. It can also assist in detecting phishing emails, generating automated responses to security alerts, or classifying log files to identify suspicious activities. In addition, the project incorporates explainable AI techniques to improve trust in LLMs 'outputs and ensures safety measures to prevent misuse of the model (e.g., adversarial prompts). By augmenting traditional security systems with LLMs, the project aims to improve the speed and accuracy of cyber threat management while reducing the burden on security analysts.

[AM-6] Development of a Browser Artifact Analysis Tool for Digital Forensics

- Suitable for CSAI students: Yes
- Project type: Software Engineering

This project aims to build a tool for analyzing browser artifacts to uncover user activities, such as web history, cookies, cached files, auto-fill data, and download records. Browser artifacts are often crucial in forensic investigations, as they can provide insights into visited websites, login sessions, timestamps, and other traces left behind during internet usage. The tool will target popular browsers (e.g., Chrome, Firefox, Edge) and extract forensic data from their databases (e.g., SQLite files) and cache directories. It will generate organized reports summarizing user activity, highlight potentially suspicious entries (e.g., visits to phishing sites or unusual patterns), and include a timeline feature to visualize user activity chronologically.

[AM-7] AI-Powered Image Recognition for Digital Forensics Evidence Extraction

- Suitable for CSAI students: Yes

- Project type: Software Engineering

This project leverages computer vision techniques to aid digital forensic investigators in analyzing images found during investigations. The system will use pre-trained deep learning models (e.g., YOLO, ResNet) to automatically identify objects, text, faces, or other relevant evidence within digital images recovered from devices or online sources. Potential applications include detecting weapons in images, identifying suspicious or illegal content (e.g., narcotics or sensitive documents), or extracting embedded text using Optical Character Recognition (OCR) for further analysis. The project will also include a filtering mechanism to prioritize relevant images flagged by the system. A key focus will be ensuring that the analysis is efficient, scalable, and explainable, with visual outputs (e.g., bounding boxes or tagged regions) highlighting areas of interest.

[AM-8] AI-Driven Social Media Evidence Collection and Analysis for Digital Forensics

- Suitable for CSAI students: Yes
- Project type: Software Engineering

The modern digital forensics process increasingly involves investigating social media platforms as they are often central to crimes, cyberbullying, misinformation, fraud, and more. This project aims to develop an AI-powered system that helps forensic investigators gather, extract, and analyze evidence from social media platforms like Twitter, Facebook, Instagram, or Reddit. The system would automate the collection of publicly available posts, images, videos, and metadata from targeted accounts, hashtags, or keywords. Using natural language processing (NLP) and AI, it would analyze the gathered data to detect suspicious activities, hate speech, misinformation, or potentially incriminating conversations. It could also detect relationships among users through social network analysis and visualize their connections.

[AM-SP] Student Proposed Project

I am happy to consider student-proposed projects, particularly those aligned with my areas of expertise. My research interests lie at the intersection of machine learning, cybersecurity, and healthcare, and I am keen to supervise projects in related domains. Specific topics I am interested in include malware detection, intrusion detection, privacy-preserving machine learning (e.g., federated learning), adversarial machine learning, digital forensics, and explainable AI applied to cybersecurity.

If you have a project idea within these areas or an innovative proposal that overlaps with these fields, feel free to discuss it with me. I am open to exploring new ideas and supporting projects that contribute to solving real-world challenges.

Boon Giin Lee's Project Ideas

General Information

In our In-the-Wild Lab, we're diving deep into two big, important areas: healthcare and education. But we're not stopping there—we're mixing in some seriously cool tech like :)

- Extended Reality (XR)
- Human–AI Interaction
- Health & Wellbeing
- Multi-modal Sensing
- Social Collaborative Computing

If you're passionate about using technology to make a real impact in people's lives—through learning or healthcare—these projects could be right up your street. Ready to explore? Here are some of the brilliant ideas we're cooking up…

These projects are led by me, in collaboration with Matthew Pike.

If you're interested in exploring these topics in healthcare or education, and want to be part of cutting-edge tech, don't hesitate to get in touch:

- Email: boon-giin.lee@nottingham.edu.cn
- Office: PMB424

Projects

[BL-1] Once Upon a Polymorphism: Storytime Meets Mixed Reality

- Suitable for CSAI students: No
- Project type: Empirical Research

What if learning tough stuff—like quantum entanglement or advanced calculus—could feel like an adventure? This project brings abstract academic concepts to life through interactive storytelling in mixed reality (MR). Think virtual characters, puzzles, and playful narrative—all designed to make complex topics easier (and more fun!) to understand.

Your mission:

- Design and build a working MR prototype for learning 1–2 tricky topics
- Test it with real students
- Collect and analyse their feedback and learning outcomes

This project is ideal for you if you have a creative mind, love problem-solving, and enjoy finding new ways to make learning stick. Coding experience and an interest in XR and education would be a big plus.

[BL-2] Go Get a Friend! An XR Social Icebreaker for Brainy Buddies

- Suitable for CSAI students: No
- Project type: Empirical Research

Let's face it—meeting new people can be awkward. This project is all about making that first hello more fun and natural using augmented or mixed reality (AR/MR) and gamification. You'll design a mobile app that helps strangers to team up, take on challenges, and win virtual rewards—all while learning and making friends.

What you'll do:

- Build a functional mobile app with AR/MR features and game mechanics
- Test it with users in real-life settings
- Analyse how well it promotes interaction, learning, and fun

This project is ideal for you if you enjoy app development, user-centred design, and the idea of using tech to bring people together in meaningful ways. Creativity and an interest in social dynamics will come in handy!

[BL-3] Rehab Rhapsody: An AI Story That Gets You Moving

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Rehabilitation can be boring—and that's a problem. This project aims to turn rehab into a daily interactive story that evolves depending on how well the patient performs their exercises. The game will use AI, motion sensors, and narrative design to increase motivation and encourage regular movement.

What you'll do:

- Create a prototype that links movement data to a personalised AI-generated storyline
- Work with real patients to test the game over a 5-day period
- Measure their engagement and analyse the data

This project is ideal for you if you love blending AI with storytelling and are excited about creating tech that supports real-world health outcomes. Interest in digital health and game design is a must!

[BL-4] The Digital Mirror: Teach an AI to Behave Like... You Know?

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This fascinating project asks: can we build an AI-driven virtual avatar that behaves like an autistic child? Using real-world autistic children's behavioural data, your goal is to train an AI to replicate body movements, facial expressions, and even patterns of communication—helping others better understand autism.

What you'll do:

- Choose and process a behavioural dataset
- Train an AI model to drive a lifelike virtual avatar
- Test it in sessions with autistic individuals and get expert feedback

This project is ideal for you if you have a strong interest in machine learning, behaviour modelling, and want to work on a project that's both technically challenging and socially meaningful.

[BL-SP] Student Proposed Project

We love original thinkers! If you've got a great project idea related to human–computer interaction, healthcare, or education, pitch it! Just bring along a 2-page proposal and come chat with me. Your idea should involve building something, testing it with users, and analysing the results.

We're looking for students who are motivated, creative, and ready to make an impact. Sound like you?

Chin Poo Lee 's Project Ideas

General Information

My research focuses on computer vision, natural language processing, and their integration through deep learning. I am particularly interested in advancing transformer-based architectures, multimodal learning, and generative models to address complex problems involving visual and textual data.

- Email: leechinpoo@outlook.com
- Office: TBC

Projects

[CL-1] Sentiment Analysis with Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project investigates advanced deep learning techniques for sentiment classification in textual data. Students will explore the use of pretrained language models, contextual embeddings, and transfer learning. Modern approaches such as prompt-based learning, adapter tuning, and contrastive representation learning may be incorporated to improve performance, especially in low-resource or domain-specific settings. Evaluation will include standard classification metrics and robustness analysis.

[CL-2] Traffic Sign Recognition with Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to develop a high-accuracy traffic sign recognition system using state-of-the-art deep learning techniques. Students will work with vision-based architectures, leveraging patch-based representations and self-attention mechanisms. Techniques such as self-supervised pretraining, data augmentation, and model compression will be considered to enhance both performance and efficiency. Emphasis will be placed on robustness to variations in lighting, occlusion, and weather conditions.

[CL-3] Gait Recognition with Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project explores the use of deep learning to identify individuals based on their walking patterns. Students will examine spatiotemporal representations and investigate models that capture sequential dynamics and structural relationships in gait data. Techniques such as self-supervised learning, temporal feature aggregation, and multi-view fusion will be explored. The project also allows experimentation with silhouette-based and pose-based input representations.

[CL-4] Plant Disease Classification with Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on developing a deep learning solution for classifying plant diseases from images. Students will use advanced image analysis techniques, including attention-based mechanisms, lightweight architectures, and automated feature learning. Additional exploration may include interpretability methods, domain generalisation, and multi-modal input integration. Emphasis will be on model accuracy, explainability, and deployment efficiency.

[CL-5] Ancient-Modern Chinese Translation with Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project involves building a neural machine translation system to translate ancient Chinese texts into modern Chinese. Students will explore sequence-to-sequence learning, attention mechanisms, and multilingual transfer techniques. Additional methods may include prompt-based learning, low-resource adaptation, and retrieval-augmented translation. Evaluation will focus on translation quality using both automatic metrics and human assessment.

[CL-SP] Student Proposed Project

I welcome student-proposed projects, particularly those aligned with current trends in artificial intelligence and data-driven technologies. Students are encouraged to bring forward their own ideas, especially if they are passionate about applying AI techniques to real-world challenges. Possible research fields include:

Computer Vision

- Natural Language Processing
- Deep Learning and Representation Learning
- Multimodal Learning (Vision + Language)

Dave Towey 's Project Ideas

General Information

The main focus of my research is currently in software testing, especially the areas of metamorphic testing (MT) and adaptive random testing (ART). I am currently particularly interested in testing so-called untestable systems (especially in areas of Big Data and Machine Learning), and autonomous vehicles.

I am also interested in exploring ways to enhance both teaching and learning, especially in areas of so-called technology-enhancement of teaching and learning, including using mixed reality (VR/AR/XR).

A final current interest for me relates to computer security.

You can contact me here:

- Email - dave.towey@nottingham.edu.cn
- Office - PMB-333

Projects

[DT-1] Testing the Untestable (& teaching how): Large Language Models

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The growing popularity of artificial intelligence (AI), especially machine learning (ML) systems, has created an incredible opportunity for innovation and evolution of our current approaches to software quality assurance and testing.

Traditional software testing typically involves executing the software under test (SUT) and observing its behaviour or output. If the behaviour or output are incorrect, then the tester knows that there is a problem with the SUT. This approach assumes a way to "know" when something is incorrect. Systems without a mechanism to "know" the correctness/incorrectness are called "untestable." Many AI/ML systems, by their very nature, are untestable.

A software testing approach called metamorphic testing (MT) has great potential for testing these untestable systems. Metamorphic exploration (ME) is a new addition to the MT literature, and involves assisting the user to become more familiar with the SUT.

Motivation: Large Language Models (LLMs), like ChatGPT, have disrupted how we engage with teaching and learning. These models can produce authentic-looking artefacts, including extensive, essay-like outputs. However, there have also been many issues identified with LLMs, including that they are not always reliable, and that they “hallucinate”(create completely fictitious contents).

Tentative Project Goals: This project has several broad goals, including developing the student's ability to engage with the growing problem of “untestable”security systems. The student will need to explore the Metamorphic Testing (MT) and Exploration (ME) literature, with LLMs as a target, develop and deploy testing strategies. A broad conception of this project includes a goal of providing some quality assurance and usage guidance for LLMs.

Open Educational Resources (OERs) are teaching, learning, and research resources that have been made available such that they can be used, shared, and modified freely. This project may also involve developing an OER.

[DT-2] Testing the Untestable (& teaching how): Autonomous Vehicles

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The growing popularity of artificial intelligence (AI), especially machine learning (ML) systems, has created an incredible opportunity for innovation and evolution of our current approaches to software quality assurance and testing.

Traditional software testing typically involves executing the software under test (SUT) and observing its behaviour or output. If the behaviour or output are incorrect, then the tester knows that there is a problem with the SUT. This approach assumes a way to “know”when something is incorrect. Systems without a mechanism to “know” the correctness/incorrectness are called “untestable.” Many AI/ML systems, by their very nature, are untestable.

A software testing approach called metamorphic testing (MT) has great potential for testing these untestable systems. Metamorphic exploration (ME) is a new addition to the MT literature, and involves assisting the user to become more familiar with the SUT.

Tentative Project Goals: This project has several broad goals, including developing the student's ability to engage with the growing problem of “untestable”AI/ML systems, identifying the implications of such systems for humanity's future, proposing solutions for these implications, and implementing and assessing the solutions.

An initial target area for this project will be Autonomous Vehicles. The selected student will need to identify an appropriate system (e.g. Baidu Apollo), negotiate/find appropriate access (e.g. a public API), and develop and deploy testing strategies with the goal of finding problems in the system.

This project may involve a replication study.

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[DT-3] Testing the Untestable (& teaching how): Virtual Production Software

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The growing popularity of artificial intelligence (AI), especially machine learning (ML) systems, has created an incredible opportunity for innovation and evolution of our current approaches to software quality assurance and testing.

Traditional software testing typically involves executing the software under test (SUT) and observing its behaviour or output. If the behaviour or output are incorrect, then the tester knows that there is a problem with the SUT. This approach assumes a way to "know" when something is incorrect. Systems without a mechanism to "know" the correctness/incorrectness are called "untestable." Many AI/ML systems, by their very nature, are untestable.

A software testing approach called metamorphic testing (MT) has great potential for testing these untestable systems. Metamorphic exploration (ME) is a new addition to the MT literature, and involves assisting the user to become more familiar with the SUT.

Tentative Project Goals: This project has several broad goals, including developing the student's ability to engage with the growing problem of "Untestable" AI/ML systems, identifying the implications of such systems for humanity's future, proposing solutions for these implications, and implementing and assessing the solutions.

An initial target area for this project will be Virtual Production (VP) software. The selected student will need to identify and explore various existing mechanisms for the software quality assurance and software testing of these systems. MT/ME will be used to explore how best to test these mechanisms. AI/ML and other techniques may be applied.

This project may involve a replication study.

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[DT-4] Testing the Untestable (& teaching how): Digitisation and Mixed Reality Software

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The growing popularity of artificial intelligence (AI), especially machine learning (ML) systems, has created an incredible opportunity for innovation and evolution of our current approaches to software quality assurance and testing.

Traditional software testing typically involves executing the software under test (SUT) and observing its behaviour or output. If the behaviour or output are incorrect, then the tester knows that there is a problem with the SUT. This approach assumes a way to “know”when something is incorrect. Systems without a mechanism to “know” the correctness/incorrectness are called “untestable.” Many AI/ML systems, by their very nature, are untestable.

A software testing approach called metamorphic testing (MT) has great potential for testing these untestable systems. Metamorphic exploration (ME) is a new addition to the MT literature, and involves assisting the user to become more familiar with the SUT.

Tentative Project Goals: This project has several broad goals, including developing the student's ability to engage with the growing problem of“Untestable”AI/ML systems, identifying the implications of such systems for humanity's future, proposing solutions for these implications, and implementing and assessing the solutions.

An initial target area for this project will be Digitisation Software and/or Mixed Reality (AR/VR/MR/XR/...) software. The selected student will need to identify and explore various existing mechanisms for the software quality assurance and software testing of these systems. MT/ME will be used to explore how best to test these mechanisms. AI/ML and other techniques may be applied.

This project may involve a replication study.

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[DT-5] Testing the Untestable (& teaching how?): Computer Security

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The growing popularity of artificial intelligence (AI), especially machine learning (ML) systems, has created an incredible opportunity for innovation and evolution of our current approaches to software quality assurance and testing.

Traditional software testing typically involves executing the software under test (SUT) and observing its behaviour or output. If the behaviour or output are incorrect, then the tester knows that there is a problem with the SUT. This approach assumes a way to "know" when something is incorrect. Systems without a mechanism to "know" the correctness/incorrectness are called "untestable." Many AI/ML systems, by their very nature, are untestable.

A software testing approach called metamorphic testing (MT) has great potential for testing these untestable systems. Metamorphic exploration (ME) is a new addition to the MT literature, and involves assisting the user to become more familiar with the SUT.

Motivation: Computer Security has been evolving as a topic, and target. By its very nature, there are many opportunities for an attacker or defender to explore and exploit systems. Metamorphic Testing principles can be applied to this area.

Tentative Project Goals: This project has several broad goals, including developing the student's ability to engage with the growing problem of "untestable" security systems. The student will need to explore the Metamorphic Testing (MT) and Exploration (ME) literature, identify an aspect of Computer Security as a target, and develop and deploy testing strategies.

This project may involve a replication study.

Open Educational Resources (OERs) are teaching, learning, and research resources that have been made available such that they can be used, shared, and modified freely. This project may also involve developing an OER.

[DT-6] Flipping Mixed and Microlearning OERs

- Suitable for CSAI students: Yes
- Project type: Software Engineering

Open Educational Resources (OERs) are teaching, learning, and research resources that have been made available such that they can be used, shared, and modified freely. Virtual Reality (VR) may refer to an interactive computer-generated experience taking place within a simulated environment (https://en.wikipedia.org/wiki/Virtual_reality). Microlearning is a holistic approach for skill-based learning and education which deals with relatively small learning units. It involves short-term-focused strategies especially designed for skill-based understanding/learning/education (https://en.wikipedia.org/wiki/Microlearning).

This project will require the student to become familiar with the background literature on Microlearning, Open Education Resources (OERs), mixed reality (virtual, augmented, ...), and classroom flipping. The student may identify a potential target experience to be recorded and turned into a VR experience. This will be recorded and then converted, with a goal of creating an OER. It will also be structured or built into microlearning-appropriate artefacts/resources. They may also be embedded in an overall structure to "flip a classroom."

[DT-SP] Student Proposed Project

I welcome any student ideas for projects, especially (although not exclusively) in areas related to my research interests. If you have any such ideas, please feel free to contact me (dave.towey@nottingham.edu.cn)

Daokun Zhang 's Project Ideas

General Information

My research interests fall into data mining and machine learning, especially on weakly supervised learning, graph machine learning and matrix factorization

If you want to work with me for FYP, please contact me by

- Email: daokun.zhang@nottingham.edu.cn
- Office: PMB443

Projects

[DZ-1] Time Series Forecasting with Missing Historical Observations

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Time series forecasting is a task that predicts future values based on sequential historical data over time. It is utilized as a key decision-making tool in various fields, such as economics and finance, supply chain management, transportation, energy, weather and healthcare. Such applications offer various opportunities, including cost reduction, increased efficiency, and enhanced competitiveness. However, the missing of some historical observations constitutes a big challenge to time series forecasting. This problem would be exacerbated for the task of high-dimensional time series forecasting. This project aims to overcome the data missing difficulty for the high-dimensional time series forecasting task through low-rank matrix factorization. The high-dimensional time series data will be modelled as a matrix, with rows corresponding to each one-dimensional time series and columns corresponding to time points. Through reconstructing the observed time series records with low-rank matrix factorization, we can learn the latent representations of different time series dimensions and the representations of different time points. With the learned latent representations, we can make predictions for the future time series observations.

[DZ-2] Time Series Forecasting with Noisy Historical Observations

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Time series forecasting is a task that predicts future values based on sequential historical data over time. It is utilized as a key decision-making tool in various fields, such as economics and finance, supply chain management, transportation, energy, weather and healthcare. Such applications offer various opportunities, including cost reduction, increased efficiency, and enhanced competitiveness. However, due to the difficulty in collecting high-fidelity data, the time series historical observations inevitably contain some noises. This project aims to overcome the data noise difficulty for the high-dimensional time series forecasting task through robust low-rank matrix factorization. The high-dimensional time series data will be modelled as a matrix, with rows corresponding to each one-dimensional time series and columns corresponding to time points. Through reconstructing the observed noisy time series records with robust low-rank matrix factorization, we can learn the latent representations of different time series dimensions and the representations of different time points. With the learned latent representations, we can make predictions for the future time series observations.

[DZ-3] Inductive Matrix Completion with Noisy Supervisions

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Inductive Matrix Completion (IMC) is a novel matrix completion formulation for relation prediction that can incorporate features associated by row and column entities. With the effective feature integration mechanism, IMC always yields significantly better relation prediction performance than the canonical matrix completion without the feature incorporation. By formulating a mapping function from the feature vectors of row and column entities to their relations, IMC can generalize to the relation prediction for entities that are not observed during the training stage. As an effective machine learning paradigm, IMC can be applied to a wide range of real-world tasks, such as recommender system, knowledge graph completion, social network link prediction, drug-disease interaction prediction, etc.

Existing IMC models measure the Mean Squared Error (MSE) between the ground-truth and predicted relations as the minimization objective for model training. However, MSE is vulnerable to noises in the relation labels, making IMC unable to perform well under the label-noise scenario. To achieve robust IMC, we will leverage the Mean Absolute Error (MAE) as the minimization objective. Due to the non-smooth property, MAE loss is difficult to be minimized by traditional gradient descent methods. In this project, we will leverage proximal gradient descent to solve the non-smooth optimization problem. Experiments on real-world datasets with relation label noises will be conducted to verify the effectiveness of the proposed robust IMC model.

[DZ-4] Inductive Matrix Completion with Incomplete Entity Features

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Inductive Matrix Completion (IMC) is a novel matrix completion formulation for relation prediction that can incorporate features associated by row and column entities. With the effective feature integration mechanism, IMC always yields significantly better relation prediction performance than the canonical matrix completion without the feature incorporation. By formulating a mapping function from the feature vectors of row and column entities to their relations, IMC can generalize to the relation prediction for entities that are not observed during the training stage. As an effective machine learning paradigm, IMC can be applied to a wide range of real-world tasks, such as recommender system, knowledge graph completion, social network link prediction, drug-disease interaction prediction, etc.

Due to the data collection difficulty or privacy/legal concerns, the features of row and column entities inevitably contain some missing values, which makes the traditional inductive matrix completion unable to accurately predict the relations between row and column entities. This project aims to leverage the low-rank matrix factorization to learn latent feature representations for row and column entities. The observations in the feature matrix of row/column entities will be reconstructed by the product between the row/column latent representations and the latent feature component representations. The feature matrix reconstruction loss is to be jointly minimized with the IMC's relation reconstruction loss. Experimental evaluation will be conducted to verify the superiority of the proposed solution over the naive attribute imputation strategies.

[DZ-SP] Student Proposed Project

The self-proposed projects on data mining and machine learning are also definitely welcome.

Fiseha B. Tesema 's Project Ideas

General Information

My research mainly focuses on computer vision, Medical Image Analysis, Deep learning, multimodal learning and fusion, audiovisual processing, and human-robot interaction.

You may contact me via:

- Email: Fiseha-Berhanu.Tesema@nottingham.edu.cn
- Office: PMB#442

Projects

[FT-1] Edge-Enhanced Dual-Stream Transformer for Small Polyp Segmentation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to improve segmentation of small polyps (<2.5% of image area) by developing a dual-stream network combining CNNs and Transformers with edge-guided feature decoupling. The model will use a CNN stream to capture local texture/shape features and a Transformer stream for long-range context, enhanced by a Laplacian Pyramid module to separate edge and body features explicitly. A boundary-aware loss function will refine edge predictions, while cross-attention fusion will dynamically integrate multi-scale features. Expected outcomes include higher accuracy (Dice/IoU) for small polyps in datasets like Kvasir-SEG and ETIS-LaribPolypDB, with real-time applicability for clinical colonoscopy. The solution addresses key limitations of current methods blurred boundaries and false positives by leveraging hybrid architectures and explicit edge modeling, as highlighted in recent surveys.

[FT-2] Dynamic Pseudo-Label Optimization for Class-Imbalanced Medical Image Analysis

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Medical image analysis often suffers from severe class imbalance, where rare anomalies are overshadowed by dominant classes, and limited labeled data restricts model performance. Existing semi-supervised methods struggle with unreliable pseudo-labels, static confidence thresholds that don ' t adapt during training, and poor generalization across different imaging modalities (e.g., 2D X-rays vs. 3D CT/MRI). This research aims to develop a robust semi-supervised framework that dynamically refines pseudo-labels based on evolving model uncertainty, integrates contrastive learning for discriminative feature representation, and extends to volumetric data through 3D-consistent pseudo-label propagation. By introducing adaptive confidence calibration, lesion-aware augmentations, and cross-modality feature learning, the proposed approach will improve detection of rare anomalies while reducing annotation dependency. The expected outcome is a scalable, data-efficient system that generalizes across diverse medical imaging tasks, enabling more reliable computer-aided diagnosis in real-world clinical settings with limited labeled data.

[FT-3] Leveraging Attention and External Knowledge for High-Performance Colon Polyp Segmentation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Accurate and efficient automatic segmentation of colon polyps is crucial for early colorectal cancer diagnosis. Building upon a prior method integrating a Pyramid Vision Transformer (PVT) encoder and a Convolutional Neural Network (CNN) decoder with a Progressive Grouped Convolutional Fusion (PGCF) module, this research outlines avenues for significant enhancement. We aim to address limitations in robustness against challenging imaging conditions, improve the handling of diverse polyp morphologies, enhance model interpretability for clinical trust, and optimize computational efficiency for potential real-time applications. This research will explore advanced attention mechanisms within the PVT-CNN framework, the incorporation of external knowledge (e.g., shape priors, multi-modal data), and the application of Explainable AI (XAI) techniques. Furthermore, we will investigate network optimization strategies and advanced data augmentation to improve robustness. The enhanced model will be rigorously evaluated on multiple datasets and compared with state-of-the-art methods. This research is expected to yield a more accurate, robust, interpretable, and efficient polyp segmentation system, paving the way for improved clinical utility in colorectal cancer screening.

[FT-4] Adaptive Class-Balanced Learning for Imbalanced Multi-Target Medical Segmentation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This study aims to address the critical challenge of scale imbalance in semi-supervised multi-target medical image segmentation, where large anatomical structures dominate model training at the expense of smaller, clinically significant targets (e.g., tumors or small organs). Current approaches struggle with rigid class representations, computational inefficiency in multi-class settings, and noisy pseudo-labels for underrepresented regions, limiting their applicability to real-world clinical datasets with diverse target sizes and frequencies. To overcome these limitations, we propose a dynamic collaborative learning framework that adaptively groups anatomically related or similarly scaled targets into shared specialist modules, optimizes their activation per image via a gating mechanism, and refines pseudo-labels through cross-specialist consensus and uncertainty-aware correction. By dynamically balancing feature learning across targets while reducing computational overhead, this method will enhance segmentation accuracy for small/rare structures and scalability to large-scale anatomical datasets (e.g., whole-body CT), ultimately bridging the gap between theoretical class imbalance solutions and practical clinical deployment. The approach will be validated on cardiac, thoracic, and multi-organ benchmarks, with metrics targeting small-structure performance (DSC, HD95) and resource efficiency (memory, inference time).

[FT-SP] Student Proposed Project

I am happy to consider project proposals from students interested in working on medical image analysis and computer vision-related topics.

Heshan Du ’s Project Ideas

General Information

My research interests are in logic, knowledge representation and reasoning, geospatial information systems, ontologies and knowledge graphs.

You may contact me by emails or drop in during my office hours.

- Email: heshan.du@nottingham.edu.cn
- Office: PMB 448

Projects

[HD-1] Validating Matches between Spatial Objects

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Geospatial data plays an essential role in many governmental, economic and social operations, such as disaster response, urban planning, outdoor or indoor navigation, tourism, etc. Establishing correspondences or matches between spatial objects described in different geospatial data sources is a critical task in integrating, enriching and updating geospatial data. Many methods have been developed to generate matches using various techniques. However, the matches generated often contain errors or mistakes, especially where location information is not complete or not accurate. This project aims to develop a method to validate the logical consistency of the matches between spatial objects.

[HD-2] Matching Historical Geospatial Data

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

A map represents locations of real-world spatial objects, such as countries, cities, roads, buildings and places, etc. Every real-world spatial object has a location. Locations of spatial objects like a city zoo, a bookstore or a university may change. The underlying data of a map is often referred to as geospatial data. A geospatial dataset contains spatial information (e.g. geometries and coordinates), temporal information (e.g. time of creation, change or disappearance) and semantic information (e.g. classifications, names, functions, etc.) of spatial objects. This project aims to develop a new algorithm for establishing correspondences (matching relations) between spatial objects represented in different geospatial datasets at different times. The algorithm will be implemented in Java (or other commonly used programming languages) and evaluated using real-world datasets and geospatial knowledge graphs.

[HD-3] Updating Location Information of Places for Tourists

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Temporal information is important for describing changes. New roads and buildings can be constructed over years. The locations of places like schools, hospitals, train stations, zoos, etc. may change over time. For example, a zoo currently in one district is planned to be moved to another district next year. This project aims to develop a method for detecting changes and updating location information of different places for tourists.

[HD-4] Reasoning with Qualitative Distance and Topological Relations

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Numerical coordinate descriptions (e.g. coordinates or geometries) of real-world entities are often hard to acquire, inherently error-prone, and probably unnecessary for most everyday tasks. Representing and reasoning with qualitative descriptions of real-world entities and their relations (e.g., near, far) are closer to human’s cognition and understanding. A spatial logic refers to a formal language interpreted over a class of structures of geometrical entities and relations. Reasoning refers to the process of deriving new facts from existing ones. Examples of topological relations are “connected”and “part of”. This project aims to develop a qualitative spatial logic which defines both qualitative distance relations and topological relations. This work involves the development of a sound and complete axiomatic system of the logic and apply the axiomatic system to reason with real-world datasets.

[HD-SP] Student Proposed Project

I am happy to consider student proposed projects in qualitative spatial logics, knowledge representation and reasoning, geospatial information systems, ontologies and knowledge graphs.

Huan Jin 's Project Ideas

General Information: Optimisation techniques, Reinforcement learning, Dynamic programming, Smart transportation, Smart logistics

- Email: huan.jin@nottingham.edu.cn
- Office: PMB438

Projects

[HJ-1] Reinforcement Learning based Dynamic AGV scheduling in smart warehouse

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to develop an automated scheduling system utilizing Automated Guided Vehicles (AGVs) to enhance workflow efficiency in smart warehouses. Addressing the operational complexity of simultaneous inbound and outbound tasks, the system will prioritize dynamic coordination of multiple AGVs performing item retrieval and delivery missions. Traditional approaches that sequentially plan individual AGV trajectories and re-plan paths upon predicting potential conflicts often result in suboptimal avoidance latency and computational redundancy. To overcome these limitations, this research proposes a reinforcement learning-based framework to enable real-time conflict-free path planning for AGV fleets. By integrating environmental perception, collaborative decision-making, and adaptive priority allocation, the system will optimize AGV trajectories through continuous interaction with warehouse dynamics, minimizing deadlock-induced delays while ensuring operational throughput. The anticipated outcomes include a scalable coordination algorithm capable of balancing task allocation efficiency with collision avoidance effectiveness under high-density logistics scenarios.

[HJ-2] 3D Mapping-based Simulator for Large-scale and Collision-free Drone Path Planning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to develop a drone flight simulator that enables collision-free drone

path planning for large-scale drone flights in complex urban high-rise environments.

This project will conduct research with following deliverables: (a) a 3D map-based simulator that visualizes and simulates drone flights in different key areas under different meteorological conditions. (b) a Reinforcement learning (RL) algorithm integrated into the 3D simulator for fast-speed, cost-effective, and collision-free path planning of a large number of drone flights. (c) AI technologies to enhance the RL algorithm. Specifically, large-language models (LLM) are embedded to improve RL's computation accuracy. In addition, the simulator will generate extensive drone flight data, enabling machine learning (ML) to

further train the RL and LLM for enhanced performance. The developed simulator and

algorithms can be widely applied in commercial fields to efficiently manage large-scale

drone flight planning.

[HJ-3] Optimization of mixed assembly line layout in garment production workshop

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The garment market is undergoing transformative changes. Traditional single-product production lines are unable to meet the current market demands of ”multiple varieties, small batches, and short cycles.” Therefore, most manufacturing enterprises are adopting modular mixed-flow production methods to replace the single production method. For example, clothing companies are combining hanging systems with intelligent unmanned handling equipment to achieve intelligent and flexible production workshops. Mixed-flow production is an extension of the traditional single-piece flow workshop production line scheduling problem, with a higher degree of flexibility and increased complexity of corresponding problems, belonging to the NP-Hard problem. The mixed-flow workshop can be reasonably optimized through resource allocation, operation sequencing, operation modes, and workstation layout to achieve efficiency improvement, energy saving, emission reduction, and cost reduction. In response to issues such as time-consuming and energy-consuming material transfer paths, uneven workload time at each workstation, long processing cycles, and waste of processing equipment resources in such production workshops, a mathematical programming model is established and solved based on constraints such as the clothing production process, limited workstation resources and processing equipment quantities, and rational production rhythm to minimize the total material transfer distance and the smoothing coefficient, and experimental verification is conducted using actual production data.

[HJ-4] Dynamic shared-charging optimization for electric vehicles and electric buses

- Suitable for CSAI students: Yes
- Project type: Empirical Research

In many developed regions, especially in urban centers and high-density residential areas, land resource constraints have made the construction of charging facilities a major challenge, and the serious shortage of charging facilities for electric private vehicles is particularly prominent. Meanwhile, existing electric bus charging stations are often idle during non-operational hours, such as at night or when buses are on duty, which results in a waste of charging resources. Based on the concept of resource sharing, opening up unused public transportation charging facilities to private vehicles during non-use hours has become a practicable solution. This strategy can not only effectively alleviate the shortage of charging piles for private cars but also bring additional revenues for public transport charging pile operators, which will help share the investment costs of charging infrastructure.

The objective of the project is : (i) propose a reinforcement learning-based electric bus charging station sharing model, which is rationally designed for the scheduling problem of sharing bus charging stations and ensures the implementation of the public transit priority principle. (ii) Considering the uncertainty of the arrival time of the buses, a customized charging reservation mechanism is developed for both buses and private cars through the reinforcement learning model to effectively manage the charging demand and optimize resource utilization.

[HJ-SP] Student Proposed Project

Proposed projects related to reinforcement learning, especially those with applications in logistics and smart transportation, are encouraged.

Heng Yu 's Project Ideas

General Information

My recent research interests fall into AI-based embedded systems, and digital systems design for AI. The projects available for FYP students in the 2025-26 cohort mainly focus on AI and related software/hardware implementation. Participating students are expected to conduct independent survey/implementation/research work, and meet me regularly for progress update, feedback, and consultancy during term and vacation time. If you are interested in a topic below, please email me directly.

- Email: heng.yu@nottingham.edu.cn
- Office: PMB426

Projects

[HY-1] 3D Reconstruction on Point Cloud Collected by an Autonomous Vehicle

- Suitable for CSAI students: Yes
- Project type: Empirical Research

With the development of autonomous vehicles, robotics, and augmented reality technology, LiDAR (Light Detection and Ranging) has become a crucial tool for obtaining three-dimensional environmental information. LiDAR emits laser beams into the surrounding environment and measures the time it takes for the reflections to return, allowing for the acquisition of highly accurate three-dimensional point cloud data. This point cloud data can be further utilized for 3D reconstruction, enabling the creation of comprehensive three-dimensional models of objects. In this project, you will first survey (and then implement one or more of) the state-of-the-art AI-based 3D reconstruction models, algorithms, datasets, etc., especially those with combined data sources from LiDAR and depth camera. Then, you are provided with a mini autonomous vehicle, equipped with the LiDAR and stereo camera. You are expected to collect the point cloud data using them, and then apply the implemented 3D reconstruction approaches to achieve reasonable outcome, in real time.

[HY-2] Spiking Neural Network for Speech Keyword Spotting

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Spiking Neural Networks (SNNs) emulate biological neural behaviour by processing information through discrete spikes, unlike traditional ANNs which use continuous activations. This event-driven mechanism enables SNNs to be particularly efficient in energy and time, making them suitable for embedded devices where efficient task execution is required. [1] have demonstrated that SNN is able for keyword spotting, and [2] shows that SNN achieved competitive results compared to ANN. The aim of this project is to re-implement [2] and apply various model miniaturization technique, such as early-exit, KD, and pruning, to improve its performance for the Noisy Speech Command Dataset.

Reference:

1. Yilmaz, Emre et al. ''Deep Convolutional Spiking Neural Networks for Keyword Spotting.''Interspeech (2020).
2. M. Dampfhoffer, et al. ''Leveraging Sparsity with Spiking Recurrent Neural Networks for Energy-Efficient Keyword Spotting.''ICASSP (2023).

[HY-3] Design and Implementation of a 2D Football Match Simulator

- Suitable for CSAI students: No
- Project type: Software Engineering

Two-Dimensional (2D) football match simulator is a system that replicates the dynamics of a football (soccer) match in a simplified two-dimensional graphical environment. In this simulator, various elements of a football match, such as player and ball movements, game events, are represented visually on a two-dimensional plane, based on predefined rules. One purpose of a 2D football match simulation can be in research and innovation. It provides an opportunity for researchers and developers to explore new technologies and methodologies in the field of simulation and artificial intelligence. Existing 2D football simulators can be find in commercial market, such as the Football Manager games, but they are not open source in nature, preventing quick data acquisition and customization. In this project, you are required to design and implement a 2D simulator from scratch. The software should simulate a football game under real-life rules and practices, facilitate well-defined input and output interfaces, and provide second-accurate data to record the events, such as player movement. Another required functionality of this simulator is 'automatic adoption', meaning that it should be able to automatically copy a match from an existing simulator, such as a 2D simulator from the Football Manager.

[HY-4] Design and Implementation of a Football Match Camera Conversion System

- Suitable for CSAI students: No

- Project type: Software Engineering

This is a sister project of [HY-3]. The intended system is to synthesize and convert a simulated football match video from the 'live camera' into a 2D representation, for which you may need to locate the players, design the geometric conversion algorithm of the pitch, map the movement from the live camera to the 2D pitch, etc. As another essential component of the system, an input creation module should be designed to adopt a simulated football match from the Football Manager game, to serve as the input for subsequent conversions.

[HY-5] Collaborated Project

I am externally collaborating with a reputable research group, in the field of electronic design automation (EDA). If you are interested and have equipped with some knowledge in this field, feel free to approach me for a discussion.

Jiawei Li's Project Ideas

General Information

Jiawei Li's research work has focused on decision making methodologies including evolutionary game theory, fuzzy logic, and hyper-heuristics, and their applications in developing strategies and algorithms.

- Email: Jiawei.li@nottingham.edu.cn
- Office: PMB426

Projects

[JL-1] Game control in the evolution of a population of intelligent agents

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The exploration of how cooperation emerges and sustains itself among self-interested agents remains a compelling and unresolved question. This project will focus on examining the evolution of cooperative behaviours within populations of living organisms or intelligent agents, utilizing the principles of evolutionary game theory. The Iterated Prisoner's Dilemma (IPD) is chosen as an ideal experimental framework due to its relevance and applicability in modelling natural processes and interactions, such as those observed in human and animal populations. In an evolutionary context, the IPD allows a population of agents to engage in repeated games, where their interactions and resultant payoffs influence their reproductive success. This dynamic simulation provides a rich environment to study the strategic complexities and emergent behaviours over generations.

The student's objectives for this project include:

1) Develop and program an IPD model that not only simulates but also visualizes the evolutionary dynamics of agent populations. This model will serve as a critical tool in observing how strategies evolve and stabilize over time.
2) Conduct a thorough analysis of the interactions between agents (strategies) and their impact on the evolutionary dynamics, focusing on how these interactions either foster or hinder cooperative behaviours.
3) Investigate evolutionary game control that adjusts evolutionary dynamics to drive the population toward cooperative or stable behaviours.

[JL-2] Simulation of economic decision making in evolutionary dynamics

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project lies within the field of computational economics, with a specific focus on bargaining games. These games model scenarios in which two or more agents negotiate over the allocation of resources. A classic example is the bargaining process between a buyer and a seller negotiating the price of a product. The goal of such negotiations is for each party to reach an agreement that maximizes their individual payoff.

In this final year project, the student will explore a market-based game in which multiple agents negotiate the exchange prices of various goods within a dynamically evolving agent population. Using an existing simulation platform, the student will track and analyze how these negotiations influence the economic environment over time, offering insights into the strategic dynamics of bargaining behavior.

Upon completion of the simulation, the student will conduct a comprehensive analysis of the results. This will help evaluate the effectiveness of different negotiation strategies and their broader implications for market dynamics.

While prior knowledge of economics, agent-based modeling, or bargaining theory would be advantageous, it is not required. The student will acquire hands-on experience in applying computational techniques to real-world economic scenarios, thereby bridging theoretical understanding with practical application.

[JL-3] Processing uncertainty in healthcare with fuzzy logic system

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Uncertainty is a pervasive challenge in healthcare, stemming from imprecise measurements, subjective symptom descriptions, and incomplete patient data. This project aims to explore how fuzzy logic systems—particularly Interval Type-2 Fuzzy Logic Systems (IT2 FLS)—can be leveraged to manage such uncertainty and improve decision-making in medical diagnosis and treatment planning.

Fuzzy logic, unlike classical binary logic, accommodates degrees of truth, making it suitable for modeling vague or imprecise information. Type-2 fuzzy logic extends this capability by incorporating a secondary layer of uncertainty through the concept of the Footprint of Uncertainty. IT2 FLS simplifies the implementation of type-2 systems by treating secondary memberships uniformly, offering a balance between computational efficiency and robust uncertainty handling.

The project will begin with a literature review of fuzzy logic applications in healthcare, highlighting successes and limitations in areas such as diagnosis support, risk assessment, and patient monitoring. It will then involve the development and simulation of IT2 FLS models using real-world medical datasets (e.g., breast cancer diagnostics or mammographic mass data). Various configurations of meet (e.g., Min, Product) and join (e.g., Max, Or, and weighted-Or) operators will be tested to evaluate their impact on classification accuracy and robustness under noisy input conditions.

Experimental results will assess the system's performance in scenarios with uncertain or conflicting inputs. Special emphasis will be placed on how operator selection influences the system's ability to maintain accuracy in the presence of input variability. The findings aim to contribute toward more interpretable and reliable decision-support tools in clinical environments, encouraging the adoption of fuzzy logic for uncertainty management in healthcare.

[JL-4] Ant colony optimization algorithm for the traveling salesman problem

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The Ant Colony Optimization algorithm (ACO) is a probabilistic method designed to solve computational problems that can be translated into finding optimal paths through graphs. This technique draws inspiration from the behaviour of real ants, where artificial ants in the model emulate the multiagent methods observed in nature. In this project, the student will explore a novel adaptation of the ACO algorithm applied to the Traveling Salesman Problem (TSP), where the interaction dynamics among the artificial ants are uniquely modified. Instead of relying solely on cooperative behaviours typically seen in ACO, this project introduces a master-slave dynamic among the ants. This new interaction model aims to enhance the efficiency and effectiveness of finding optimal solutions. The student's responsibilities will include developing software that not only implements this revised ACO algorithm but also assesses its performance through rigorous evaluation. Additionally, the software will feature visualization capabilities, allowing for the graphical representation of the solutions to the TSP. This will aid in better understanding and analysing the paths generated by the algorithm, providing valuable insights into the impact of the new ant interaction dynamics on the problem-solving process.

[JL-SP] Student Proposed Project

I am happy to consider student project proposals in the following topics:

- Computational intelligence (in decision making or optimization problems)

- AI in games

If you would like to do one of these topics with me then send me an email describing your idea.

Jianfeng Ren 's Project Ideas

General Information

My main research areas are computer vision, image/video processing, statistical pattern recognition, deep learning, human-computer interaction and radar target recognition. I have gradually shifted my research interests in deep learning, especially visual reasoning and understanding. The following projects are related to deep-learning techniques.

- Email: Jianfeng.ren@nottingham.edu.cn
- Office: PMB445

Projects

[JR-1] Visually Grounded Commonsense Knowledge Acquisition

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Work type distribution: 70% research, 30% development

Pre-requisites: machine learning, good knowledge on deep learning, natural language processing, research experience

Previous approaches have attempted to extract commonsense knowledge from plain text or pre-trained language models (PLMs). However, commonsense is rarely reported in text, and commonsense in PLMs suffers from low consistency and significant reporting bias. There is also widespread doubt whether learning purely from the surface text forms can lead to real understanding of commonsense.

In this project, the student is expected to explore various techniques for acquiring visually grounded commonsense knowledge given a set of images. The student is expected to conduct pioneer research on this topic, develop new techniques, and outperform the state-of-the-art algorithm in this research area after closely cooperation with the supervisor. The student is expected to produce a research paper along this direction at the end of this project.

[JR-2] Visual Understanding/Reasoning Using Deep Learning

- Suitable for CSAI students: Yes

- Project type: Empirical Research

Work type distribution: 70% research, 30% development

Pre-requisites: machine learning, good knowledge on deep learning, visual understand/reasoning, research experience

In recent years, deep learning techniques have advanced significantly, which moves from recognizing single objects, to understand the image scene and conduct reasoning based on the understanding of the image scene. A lot of research efforts have been devoted to developing a visual reasoning system that can not only visually recognize objects of

scenes, but also conduct logical reasoning based on the perceived visual information. Typical applications for visual reasoning/understanding include Visual Question Answering, Raven's Progressive Matrices, Puzzle Reassembly, etc.

In this project, the student is expected to explore various techniques for image understanding/reasoning. The student is expected to conduct pioneer research on this topic, develop new techniques, and outperform the state-of-the-art algorithm after closely cooperation with the supervisor. The student is expected to produce a research paper along this direction at the end of this project.

[JR-3] Sparse Representation and Its Applications in Machine Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Work type distribution: 70% research, 30% development

Pre-requisites: machine learning, good knowledge on dictionary learning, sparse representation, research experience

This is a research-extensive project. Sparse representation has been widely used in many fields. In this project, the student is expected to explore the feasibility of building a general framework to extend the current sparse representation for better classification performance. The student needs to implement existing state-of-the-art sparse representation models and build a new framework. Then, extensive experiments on a large number of datasets need to be conducted to evaluate the performance gain of the proposed method. At the end of this project, the student is expected to co-author a research paper as the output of this project.

The applicant is preferred to have excellent mathematical modelling skills and outstanding programming skills to successfully deliver this project.

[JR-4] Data Discretization and Feature Selection/Weighting for naive Bayes Classifier

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Work type distribution: 70% research, 30% development

Pre-requisites: machine learning, data discretization

This is a research-extensive project, which is an extension of our recent publication on naive Bayes classifier. The Bayesian classification framework has been widely used in many fields, but the covariance matrix is usually difficult to estimate reliably. In order to alleviate this problem, many naive Bayes (NB) methods with good performance have been developed. However, the assumption of conditional independence among attributes in NB is rarely established in reality. Various attribute weighting schemes have been developed to solve this problem. Among them, attribute-weighted Naive Bayes (CAWNB) has recently achieved good performance by using classification feedback to optimize the attribute weight of each category. However, the derived model may be over-fit to the training data set, especially when the data set is not sufficient to train a model with good generalization performance. In addition, data discretization and feature selection/weighting should be jointly considered in order to maximize the discrimination power of the classifier.

In this project, the student is expected to explore the feasibility of building a general framework to jointly consider the data discretization problem together with the feature selection/weighting for better classification performance. The student needs to implement an existing state-of-the-art naive Bayes classifier, or build a new framework. Then, extensive experiments on a large number of datasets need to be conducted in order to evaluate the performance gain of the proposed method. At the end of this project, the student is expected to co-author a research paper as the output of this project.

[JR-SP] Student Proposed Project

Detail here if you 're happy to consider student proposed projects and the subject areas you are interested in supervising.

The projects listed above are merely some examples which are suitable for final year project. The students are welcome to propose your own project ideas. For my own preference, the projects should have some research values, and should be able to complete within one year. If you have some good ideas and you are not sure whether it is suitable for DT project, you may contact me through email: ***jianfeng.ren@nottingham.edu.cn***. The preferred areas (but not limited to) are:

- Image processing
- Computer vision

- Deep learning
- Radar/audio/image recognition

Kian Ming Lim 's Project Ideas

General Information

My research interests include deep learning, computer vision, natural language processing, few-shot learning, text-to-image generation, and audio/signal processing. My expertise lies in leveraging deep learning techniques to solve real-world problems in these domains. I am particularly focused on developing intelligent models that can learn from limited data, handle multimodal inputs, and perform robust analysis across visual, textual, and audio signals.

You may contact me via:

- Email: kian-ming.lim@nottingham.edu.cn
- Office: PMB424

Projects

[KL-1] Deep Learning-based Medical Image Classification for Disease Diagnosis

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Medical image classification plays a vital role in modern healthcare by enabling early and accurate diagnosis of diseases such as cancer, pneumonia, and neurological disorders. Deep learning has emerged as a powerful tool for automating this process, offering the potential to reduce human error and enhance diagnostic efficiency. However, challenges such as limited annotated datasets, class imbalance, and the need for model interpretability remain significant obstacles in deploying reliable AI solutions. This project focuses on developing and evaluating a deep learning-based system for medical image classification using datasets from modalities such as X-rays, MRI, or CT scans. The work will involve exploring advanced architectures including convolutional neural networks, vision transformers, or hybrid models, and applying techniques such as data augmentation, transfer learning, attention mechanisms and large models to improve classification performance. The expected outcome is a robust deep learning model capable of accurately classifying medical images, potentially surpassing current benchmarks. The final deliverable may also include a research paper documenting the methodology, experimental setup, results, and key findings.

[KL-2] Deep Learning for Hyperspectral Image Classification in Remote Sensing

- Suitable for CSAI students: Yes

- Project type: Empirical Research

Hyperspectral imaging captures detailed spectral information across numerous narrow bands, making it a powerful tool in remote sensing applications such as land cover classification, environmental monitoring, agriculture, and mineral exploration. However, the high dimensionality of hyperspectral data, limited labeled samples, and the need to effectively integrate spectral and spatial features pose significant challenges for traditional classification methods. This project focuses on building a deep learning-based framework for hyperspectral image classification. The work will involve investigating architectures capable of learning both spectral and spatial patterns, such as 3D convolutional neural networks, hybrid CNN-RNN models, transformer-based approaches, or large models. Techniques like dimensionality reduction, data augmentation, and transfer learning will also be considered to improve model generalization and training efficiency. The goal is to develop a high-performance classification model using benchmark hyperspectral datasets, delivering accurate and interpretable results for real-world remote sensing tasks. The final deliverables will include a comprehensive research paper detailing the model design, implementation, experimental results, and key findings.

[KL-3] Deep Learning for Acoustic Event Classification in Real-World Environments

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Acoustic event classification aims to identify and categorize sounds from real-world environments, such as footsteps, sirens, speech, door knocks, or alarms. It plays a crucial role in applications such as smart surveillance, ambient assisted living, urban monitoring, and intelligent personal assistants. Deep learning has emerged as a powerful approach for modeling the temporal and spectral features of acoustic signals, yet challenges remain due to background noise, overlapping events, and the diversity of sound sources. This project focuses on developing a deep learning-based system for classifying acoustic events from audio recordings. The work includes transforming raw audio into suitable input formats such as spectrograms or mel-frequency cepstral coefficients, followed by the design and training of models such as convolutional neural networks, recurrent neural networks, or transformer-based architectures. Techniques like large models, transfer learning, and noise-robust preprocessing will also be explored to improve classification accuracy in real-world conditions. The objective is to build an effective and generalizable acoustic event classification system using public datasets. The outcome of the project will include either a functional prototype or a robust evaluation framework, accompanied by a detailed research report that documents the methodology, experimental setup, results, and key insights.

[KL-4] Customer Churn Prediction Using Deep Learning for Business Intelligence

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Customer churn prediction is essential for businesses seeking to retain customers and ensure long-term profitability. By identifying individuals who are likely to stop using a service or product, companies can take proactive steps to improve engagement and reduce attrition. Traditional statistical models often struggle to capture the complex behavioral patterns and interactions present in customer data. In contrast, deep learning offers a powerful solution by automatically learning representations from diverse and high-dimensional data sources. This project focuses on developing a deep learning-based approach to predict customer churn using historical behavior, transaction records, demographic information, and service usage patterns. The investigation will explore models such as fully connected neural networks, recurrent neural networks for temporal data, transformer architectures, and large-scale models to improve both interpretability and performance. Additionally, techniques such as feature engineering, class balancing, and model explainability will be employed to enhance insight and transparency. The final outcome will be a comprehensive research paper that documents the problem formulation, data preprocessing, model development, experimental results, and key findings on the effectiveness of deep learning for customer churn prediction.

[KL-5] Multi-Disease Prediction from Health Indicators Using Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Accurate and early identification of multiple diseases based on patient health indicators is essential for effective diagnosis and timely intervention. Health records often contain structured data such as vital signs, lab test results, demographic details, and medical history, which can be leveraged to predict the presence of various conditions. Traditional classification models may fall short when dealing with the complex, multi-label nature of such tasks. Deep learning provides a scalable and flexible approach for modeling high-dimensional, heterogeneous health data. This project aims to develop a deep learning-based model to perform multi-disease classification using structured health indicators. The work involves investigating models capable of multi-label classification, such as fully connected neural networks, attention-based models, and large-scale architectures that can capture feature interactions and correlations among diseases. Techniques like data normalization, feature selection, and class imbalance handling will also be explored to improve model performance and generalization. The outcome of the project will be a detailed research paper that presents the methodology, data preprocessing steps, model design, evaluation metrics, and key findings on the use of deep learning for multi-disease prediction based on health indicators.

[KL-SP] Student Proposed Project

I am happy to supervise student-proposed projects that align with my research interests. Students are encouraged to propose project ideas that reflect their own interests and career goals. Suitable topics may fall within the broad areas of artificial intelligence, deep learning, computer vision, natural language processing, or data analytics. Projects should demonstrate a clear research or application focus, with well-defined objectives, relevant datasets, and achievable outcomes. Proposals will be evaluated based on their originality, feasibility, and relevance to current technological challenges. Please feel free to contact me via email for further discussion.

Matthew Pike's Project Ideas

General Information

My projects this year will be based in our HCI "Research in the Wild" Lab. The lab focuses on real-world applications of research, emphasising ecological validity by testing concepts with real people in realistic environments. Our main areas of focus are AI, Education, and Healthcare, often utilising XR and sensor technologies. Students working on the projects listed below will collaborate within the lab, providing regular group-based weekly updates and offering mutual support, supervised by myself and Boon Giin Lee.

- Email: matthew.pike@nottingham.edu.cn
- Office: PMB435

Projects

[MP-1] Mental Workload Classification System (MWCS)

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Overview The aim of this project is to ingest quantitative, brain-derived data from a brain sensing device such as functional Near-Infrared Spectroscopy (fNIRS) or Electroencephalography (EEG) and classify the wearer's current level of mental workload (MWL)—for example, as low, normal, or high. The intention is to develop a robust, real-time classification system that can serve as an intelligent input for other adaptive systems (e.g., user interface adjustments, alert systems, or training environments).

Background Recent research has highlighted the potential of neuroimaging techniques such as fNIRS and EEG for assessing cognitive states like MWL. This project builds on foundational work in neuroergonomics and adaptive system design, including:

- Classification Algorithm for fNIRS-based Brain Signals Using Convolutional Neural Network with Spatiotemporal Feature Extraction Mechanism
- Benchmarking framework for machine learning classification from fNIRS data

Objectives Students undertaking this project will be expected to:

- Collect a labelled dataset using a brain sensing device (e.g. fNIRS or EEG) during tasks of varying cognitive load (e.g. N-back tasks, Stroop test).
- Pre-process and extract features from the raw neurophysiological signals using appropriate signal processing techniques.
- Train and evaluate a machine learning model to classify mental workload levels.
- Test the trained model in real-time, using streaming data from the brain sensing device to validate its performance in dynamic conditions.

[MP-2] Robust Breathing Phase Detection in Everyday Life Using Mobile Sensors

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Overview This project aims to detect and classify breathing phases—specifically inhalation and exhalation—using a standard smartphone placed on the user's chest, such as in a shirt pocket. The goal is to enable real-time monitoring in unconstrained, everyday environments and to use this information to provide adaptive haptic feedback (e.g., vibration) that promotes healthier breathing patterns.

Background Breathing phase detection has traditionally relied on dedicated wearables or laboratory equipment. While these methods offer accuracy, they are often impractical for everyday use. Mobile phones, however, are equipped with accelerometers and gyroscopes that can capture subtle chest movements associated with respiration.

The challenge lies in developing a system that can robustly detect breathing phases not just in static conditions, but during normal activities—such as desk work, casual walking, or using a computer—where motion and noise are common.

This research aligns with growing interest in mobile health (mHealth) and the quantified self movement. The application potential spans mindfulness training, stress reduction, and even cognitive performance enhancement through guided oxygenation practices.

Related works include:

- BreathTrack
- Respiration Rate Estimation Based on Independent Component Analysis of Accelerometer Data: Pilot Single-Arm Intervention Study

Objectives

- Collect sensor data (e.g., accelerometer and gyroscope) from a mobile phone placed on the user's chest during various activities.
- Pre-process the data and extract features relevant to respiratory motion.
- Train and evaluate a classification model to detect inhalation and exhalation phases.
- Test the robustness of the system in both static and dynamic everyday scenarios.
- Develop a simple prototype that uses haptic feedback to nudge users towards healthier breathing rhythms.
- (Optional extension) Explore adaptive feedback strategies tailored to user goals—e.g., relaxation, focus, or energy.

[MP-3] Adapting Search Engine Interfaces for User's Cognitive Styles

Overview
This project aims to implement a cognitive-style-tolerant search engine, building upon the current Google search engine, to better support users with diverse information needs, particularly for exploratory search tasks.

Background
There is a growing demand for online search capabilities that extend beyond simple look-up tasks to support more complex exploratory search needs. However, designing search engines that effectively cater to the higher levels of personalization required for exploratory search remains challenging due to a lack of clear guidelines. Research indicates that personal differences, especially cognitive style, significantly influence users' search interface preferences and behaviours. Related works include:

- Cognitive styles and search engine preferences: Field dependence/independence vs holism/serialism

- Exploring the impact of verbal-imagery cognitive style on web search behaviour and mental workload

Objectives
The project objectives are to:

1. Implement a search user interface designed to accommodate different cognitive styles.
2. Improve the query suggestion mechanism and search result display to better support users with varying cognitive styles.

[MP-4] SimSocAI: An Evolving Social Network for Autonomous AI Agents

- Suitable for CSAI students: Yes
- Project type: Software Engineering

Overview SimSocAI provides a digital environment where AI agents ('users'), each equipped with distinct personalities, goals, and memory, can interact, adapt, and evolve. These users will be capable of posting updates, replying to one another, following trends, and even engaging in debates, such as arguing over football results. Significantly, the environment will incorporate external stimuli —news headlines, sports scores, or miscellaneous facts and quotes—delivered through dedicated AI channels representing public data sources. The project aims to explore how artificial agents interact and develop within a simulated society, responding both to real-world information and to each other, all presented through the familiar interface of a social network. SimSocAI should be a looking glass for us to look into an simulated AI social network.

Background Much current AI research focuses on individual agent capabilities or interactions within limited, task-specific contexts. Understanding how complex, often unpredictable social dynamics emerge within populations of interacting, learning agents presents a significant challenge. Related works include:

- Generative Agents: Interactive Simulacra of Human Behavior
- The Spread of Behavior in an Online Social Network Experiment

Objectives

- Develop a robust platform for simulating an AI social network.
- Implement the core agent architecture (LLM integration, memory, personality).

- Construct the social network backend and frontend interface.
- Integrate external data feeds via dedicated data source agents.

Ning Xue 's Project Ideas

General Information

His current research interests focus on computational intelligence, data-driven optimization approaches, operations research, evolutionary algorithms, modeling, scheduling, and optimization in transportation and hospitality sectors, as well as explainable AI.

- Email: ning.xue@nottingham.edu.cn
- Office: PMB435

Projects

[NX-1] Joint optimization of multi-intelligent distribution vehicle routes for urban shared freight transportation

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

This project addresses the joint optimization problem of multi-agent vehicle routing in urban shared freight systems, aiming to simultaneously optimize task allocation, route sequencing, economic costs, delivery efficiency, and carbon emissions. We propose a novel three-stage hierarchical hybrid optimization framework that integrates integer programming, multi-traveler path optimization, and deep reinforcement learning. In the first stage, we formulate the scheduling problem as a constrained Vehicle Routing Problem (VRP) and employ integer linear programming (ILP) solvers (e.g., Gurobi or Google OR-Tools) for initial task-vehicle allocation, ensuring compliance with practical constraints including vehicle capacity, service time windows, and shared warehouse transfers. The second stage optimizes local path costs for each vehicle using the Lin-Kernighan-Helsgaun (LKH) heuristic algorithm to solve the Traveling Salesman Problem (TSP). In the final stage, a Proximal Policy Optimization (PPO)-based reinforcement learning model performs real-time path fine-tuning, adapting to dynamic urban conditions such as traffic fluctuations, road congestion, and energy constraints. This RL agent processes a state vector encompassing route, traffic, and vehicle status data, while its reward function balances delivery time, emissions, and shared resource utilization.

[NX-2] AI-Optimized Last-Mile Deliveries in Urban Environments

- Suitable for CSAI students: Yes

- Project type: Theoretical Research

We propose an intelligent scheduling framework that combines Graph Convolutional Networks (GCN) and Multi-Agent Reinforcement Learning (MARL) to dynamically optimize collaborative distribution strategies across heterogeneous fleets, including unmanned ground vehicles, drones, and shared smart lockers. The system models the urban logistics network as a dynamic graph structure, where nodes represent distribution points (warehouses, smart cabinets, and delivery locations) and edges encode real-time traffic conditions and energy costs. Spatial features extracted through GCN processing are integrated with MARL to develop adaptive scheduling policies that simultaneously address three critical objectives: minimizing delivery delays, reducing carbon emissions, and alleviating road congestion. A novel multi-objective reward function aligns the system's optimization with sustainability goals, while the hierarchical architecture enables efficient coordination between different vehicle types with varying operational constraints. The framework is designed to process real-time urban data streams, enabling dynamic route adjustments in response to changing traffic patterns and demand fluctuations.

[NX-3] Graph Assisted Offline-Online Deep Reinforcement Learning for Dynamic Workflow Scheduling

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

We propose a novel offline-online deep reinforcement learning framework that combines graph-based representation with hybrid optimization techniques to address complex workflow scheduling in dynamic environments. Our approach is particularly designed for task-dependent, resource-constrained scenarios across urban logistics, intelligent manufacturing, and multi-AGV systems. The framework begins by modeling workflows as directed graphs, where nodes represent tasks and edges capture priority dependencies. We integrate task attributes (including execution time, priority, and resource requirements) with structural features extracted through Graph Neural Networks (GNNs) to create comprehensive task representations. The offline phase employs metaheuristic algorithms (Simulated Annealing and Genetic Algorithms) to generate high-quality scheduling solutions from historical data, which serve as expert demonstrations for behavioral cloning and RL policy initialization. For online operation, we implement an asynchronous Advantage Actor-Critic (A3C) architecture that processes real-time system states (encoded via GNNs) including task graph structure, resource availability, and completion status. This enables dynamic policy updates in response to disturbances such as new task arrivals or resource failures. The reward function incorporates multiple optimization objectives spanning makespan, resource utilization, task delays, and interruption costs. The framework's parallel training design ensures efficient handling of concurrent scheduling requests while maintaining system stability.

[NX-4] Unsupervised Learning for Multi-Vehicle VRP with Charging and Time Windows

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

We propose a novel unsupervised graph learning framework to address the complex challenges of multi-vehicle routing problems with charging and time window constraints (VRP-C-TW). Our approach eliminates the dependence on labeled training data by integrating Graph Neural Networks (GNNs) with a self-supervised optimization paradigm. The urban distribution network is modeled as a graph G=(V,E), where nodes represent customers, warehouses, and charging stations, while edges encode feasible routes with associated distance, energy consumption, and service time attributes. The GNN architecture generates edge probability heatmaps that indicate optimal path selection likelihoods, trained through a custom unsupervised loss function combining: (1) path cost-weighted optimization and (2) constraint violation penalties for charging requirements and time windows. These heatmaps subsequently guide heuristic local search algorithms (e.g., 2-opt, ALNS) to construct complete vehicle schedules. To specifically address electric vehicle constraints, we incorporate a dynamic multi-stage charging insertion strategy during path optimization. The framework's key innovation lies in its end-to-end unsupervised learning capability, which avoids the need for pre-computed optimal solutions while handling multiple real-world constraints natively through the graph representation and loss formulation. This approach is particularly designed for urban shared freight systems requiring scalable solutions with minimal deployment overhead.

Pushpendu Kar ’s Project Ideas

General Information

My research interests include Wireless Sensor Networks, Internet of Things, Content-Centric Networking, Blockchain, Machine Learning, Artificial Intelligence, and BigData.

Students can contact me by email or in my office to discuss the project:

- Email: pushpendu.kar@nottingham.edu.cn
- Office: PMB448

Projects

[PK-1] Secure and Intelligent Document Verification System Using Blockchain and AI

- Suitable for CSAI students: Yes
- Project type: Empirical Research

In today's digital age, document fraud and forgery have become widespread challenges, affecting individuals, businesses, and governments. Traditional document verification techniques often rely on centralized systems, leaving them vulnerable to tampering, inefficiencies, and user privacy risks. This project proposes an innovative solution by integrating blockchain and artificial intelligence (AI) to create a secure and intelligent document verification system.

The blockchain component ensures immutable storage and decentralized verification of documents, protecting against unauthorized changes and providing transparency. Smart contracts are used to automate document validation processes and transactions securely. On the other hand, AI-driven models are employed to identify fraudulent documents, verify authenticity, and provide intelligent recommendations in real-time.

The proposed system will allow users and organizations to upload, verify, and track the authenticity of critical documents such as certificates, identity proofs, contracts, and legal papers. Blockchain ensures tamper-proof storage and ownership tracking, while AI enhances accuracy in detecting forgery patterns. The system will demonstrate its capabilities through a prototype platform with a user-friendly interface, showcasing practical use cases in education, healthcare, and legal industries.

This integration of blockchain ’s transparency and security with AI ’s cognitive intelligence addresses key challenges in document management, offering a future-proof solution for digital trust. It holds enormous potential for wider applications, including digital identity management and fraud prevention across various sectors.

[PK-2] Development of an Efficient and Low-Power AI- Specific Consensus Mechanism for Blockchain.

- Suitable for CSAI students: Yes
- Project type: Empirical Research

In the practical transaction, the blockchain consensus protocol consumes a large amount of computing resources and power, leading to low system throughput and long system latency. However, the blockchain requires higher operational and more compatible capabilities for the existing platforms. Designing an interaction mechanism to aggregate and utilize the group intelligence of distributed consensus nodes is an important issue that the blockchain needs to deal with. Although the foundation of blockchain technology is relatively mature, there are still many problems to be solved in protocol improvements. The target is how to construct the consensus mechanisms to improve system throughput. For example, the consensus algorithm of main nodes selection from a few trusted nodes, the asynchronous consensus algorithm of increasing the probability of high correctness, the consensus algorithm of reduction of network broadcasts based on specific security premise, the consensus algorithm based on trusted hardware, and the consensus algorithm combining the advantages of PoW and PBFT. Until now, we have a number of protocols specific to the standards and features of different blockchain platforms. The research on consensus mechanisms has been compared and analyzed from different angles. The common goal is to seek better protocols that can achieve higher levels of energy and cost saving, tolerated power of adversary, identity and execution, scalability, security, privacy, etc. Existing consensus protocols consider network and middleware layers of blockchain systems by enabling different proof of X protocols. A large plethora of research opportunities are available for future researchers to explore if the application-level consensus protocols could be designed considering proofs based on the quality of learning models, efficient search strategies, quality and provenance of data, and quality of optimization.

[PK-3] AI-Powered Adaptive Content Delivery in Data-Centric Networks for Next-Generation Applications

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The exponential growth of data-centric services, such as augmented reality (AR), virtual reality (VR), IoT, edge computing, and content streaming, has brought new complexities to traditional network infrastructure. Data-Centric Networking (DCN) addresses these challenges by focusing on data itself rather than specific endpoints. However, efficiently managing content delivery, maintaining Quality of Service (QoS), and reducing latency in DCNs requires dynamic, intelligent systems capable of adapting to user demands and network conditions. This research explores the integration of Artificial Intelligence (AI) into DCN architecture to create an adaptive, self-optimizing content delivery network tailored for next-generation applications.

In this project, AI models will be utilized to actively predict content demand patterns based on user behavior, forecast network congestion, and optimize caching and routing strategies for delivering data. The system will use deep reinforcement learning to continuously learn and adapt routing paths, ensuring minimal latency and improved throughput. Additionally, predictive AI algorithms will identify popular content ahead of demand spikes, enabling the proactive placement of data in edge nodes and caches. Anomaly detection models will monitor and address network inefficiencies, such as bottlenecks, packet loss, and underutilized resources, in real time.

The project will culminate in a simulated prototype demonstrating intelligent, scalable content delivery in a DCN environment, and will validate its effectiveness in scenarios such as immersive gaming, IoT data exchange, real-time video streaming, and mission-critical applications. By blending AI 's foresight and adaptability with DCN 's focus on content, this innovative solution aims to redefine content delivery in modern networks, paving the way for next-generation, data-intensive services and applications.

[PK-4] AI and Blockchain-Powered Early Disease Detection Network

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project combines AI-powered predictive analytics and blockchain technology to create an early disease detection network for hospitals or local healthcare providers. Patient health records, wearable device data, and diagnostic results are stored in a blockchain to ensure privacy while facilitating secure sharing across healthcare providers. AI algorithms analyze health data to predict and detect diseases such as diabetes, cancer, or cardiovascular issues at early stages, improving chances of recovery. Blockchain enforces data ownership and consent, allowing patients to control how their data is shared with researchers. This system aims to improve diagnostic accuracy and enable proactive healthcare interventions.

[PK-5] Decentralized AI Model Sharing Platform with Blockchain Security

- Suitable for CSAI students: Yes

- Project type: Empirical Research

This project proposes a platform that enables secure sharing and collaboration on AI models using blockchain technology. The system aims to address challenges in model distribution, ownership, and integrity. Blockchain serves as the backbone for storing metadata about AI models, ensuring immutable records of their creation, updates, and ownership. AI models can be uploaded, shared, or sold through the platform, with smart contracts automating licensing agreements and royalty payments. Additionally, the blockchain ledger tracks usage, ensuring ethical application of AI while mitigating risks such as unauthorized modifications or intellectual property theft. This system caters to AI researchers, businesses, and developers who seek secure and transparent collaboration on AI technologies.

[PK-SP] Student Proposed Project

I welcome any student ideas for projects in the computer science and engineering domain especially (not limited to) in research areas of my interests. Students with new ideas will come to me with an abstract to discuss the ideas.

Qiao Lin 's Project Ideas

General Information

My research interests are in Trustworthy AI, Medical Image Segmentation, Uncertainty Analysis of AI Models, Fuzzy Logic, and Hallucinations Issues of Large Models. If you are interested in the following projects, feel free to schedule a meeting with me either online or offline.

• Email: qiao.lin@nottingham.edu.cn

• Office: PMB123

Projects

[QL-1] Uncertainty Estimation for Semantic Segmentation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Semantic segmentation essentially refers to pixel-level classification, the objective of which is to classify each pixel of the input image into different categories and then segment the given image into various useful and meaningful regions based on the pixel classification results. The general framework of semantic segmentation models consists of two parts: (1) image convolution and down-sampling are utilized to extract spatial and channel information of given images; (2) up-sampling or deconvolution is applied to resize the output prediction to the same size as the training image label. Model uncertainty and data uncertainty exist in semantic segmentation. Accurate measurement of uncertainty benefits the improvement of segmentation performance and model reliability. This research aims to propose an uncertainty estimation algorithm for semantic segmentation.

[QL-2] Prompt-engineering-based Hallucination Estimation for large language models

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Hallucination of Large Language Models (LLMs) refers to the phenomenon where the model generates information that is non-existent or false. This hallucination arises due to the limitations and biases inherent in the model, causing it to produce content that may not align with actual facts or common sense. Hallucination in LLMs is a complex issue that is difficult to completely eliminate. Since the models are trained on vast amounts of data, they may inadvertently learn incorrect or misleading information and incorporate it into their generated text. Additionally, the models' limitations in understanding and processing natural language can lead to inaccuracies or misunderstandings of certain concepts or facts. This research aims to propose a Prompt-engineering-based hallucination estimation algorithm of LLMs to improve the reliability of LLMs.

[QL-3] Image classification for multiphase flow systems

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Multiphase flow systems are widely used in chemical, pharmaceutical, and environmental industries. However, the multiphase flow is usually highly dynamic and very complex, hindering the development of multiphase flow theory and the optimization of multiphase flow systems. Through the experiments in the pilot-scale systems, massive experimental data, including hydrodynamics and system performances has been accumulated. For this research, the inputs are images captured with a high-speed camera, the outputs are operating conditions, particle properties. **Requirement**: after the training, the artificial neural networks (ANN) shall be able to well correlate the inputs with outputs. If we input an image or several images, the trained ANN shall be able to recognize the operating conditions and particle properties. **Application**: this project can be used to monitor the multiphase flow system. If there is any deviation from the desired operations, ANN can be used to recognize and inform the possible circumstances.

[QL-4] Interpretation of pressure drops for multiphase flow systems

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Multiphase flow systems are widely used in chemical, pharmaceutical, and environmental industries. However, the multiphase flow is usually highly dynamic and very complex, hindering the development of multiphase flow theory and the optimization of multiphase flow systems. For this research, you will use AI-related algorithms to handle some practical multiphase flow systems issues. **Inputs**: pressure drop signals (average value, standard deviation, energy information obtained from wavelet analysis), axial locations **Output**: operating conditions, particle properties, radial distributions of solids holdup, radial distributions of heat transfer coefficients. **Requirement**: after the training, the artificial neural networks (ANN) shall be able to well correlate the inputs with outputs. If we input the pressure drop signal or information, the trained ANN shall be able to recognize the operating conditions, particle properties, radial distributions of solids holdup, radial distributions of heat transfer coefficients.

[QL-SP] Student Proposed Project

Students are welcomed to propose their project ideas and the subject area I am interested in supervising is trustworthy AI. Feel free to email me and schedule a meeting to discuss.

Qian Zhang 's Project Ideas

General Information

I am interested in the intersection of large language models, computer vision, and human-computer interaction. My research focuses on building context-aware, multimodal systems for action understanding, quality assessment, and structured feedback generation—especially in domains such as rehabilitation, smart manufacturing, and education.

For the research-based project, you will follow a research procedure. You need to focus on algorithm design, analysis, and evaluation. For the application-based project, you need to follow the software engineering procedure. There are no special requirements for all my projects. For project discussion and selection, please send me an email to arrange a meeting if you are interested in my projects. I will arrange only one session for each project. If you get the project, you are required to manage your own learning activities actively and consciously. You need to make full use of 400 hours on self-learning, design, coding, and writing. Of course, proper time management and an early start are preferred. For project supervision, I ' d like to use the result-target format. We will discuss your previous result and set up your next target during each meeting, and we will not waste time if no progress has achieved.

- Email: qian.zhang@nottingham.edu.cn
- Office: PMB438

Projects

[QZ-1] Context-Aware Action Recognition from Multimodal Video

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on designing a system that can recognize and label complex human actions in video using multiple data sources such as RGB video, pose estimation (skeleton data), and textual task descriptions. The student will explore time-aware fusion techniques (e.g., Transformer-based encoders) to integrate visual and textual features, and evaluate performance on existing action recognition benchmarks with added temporal and contextual constraints.

[QZ-2] Generating Structured Natural Language Feedback for Human Motion Quality

- Suitable for CSAI students: No

- Project type: Software Engineering

This project involves implementing a feedback generation module that translates low-level performance metrics (e.g., pose deviation, timing inconsistency) into natural language suggestions for improvement. The student will design templates and control signals to interface with a large language model (e.g., GPT or LLaMA variants) and validate the quality of generated feedback using a rehabilitation or fitness-oriented dataset.

[QZ-3] Few-Shot Learning for Multimodal Action Understanding

- Suitable for CSAI students: Yes
- Project type: Research

This project investigates few-shot or low-data learning techniques for video action recognition using multimodal pseudo-labeling and distillation. The student will implement and evaluate knowledge distillation strategies at the label, representation, and attention levels—transferring knowledge from a pretrained LLM or VLM to a lightweight student model using limited annotated data.

[QZ-4] Pyramid-Based Multilevel Intention Modeling for Task-Oriented Video Analysis

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on building a pyramid-style intention recognition system to infer atomic, operational, and task-level user goals from a video sequence. The student will combine GNNs and Transformers to model user behavior hierarchies and test the model on real-world task datasets, such as guided rehabilitation exercises or assembly tasks.

[QZ-5] Multimodal Benchmarking for Explainable Human Action Evaluation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to design and evaluate a benchmark tool that assesses human action sequences across multiple modalities (video, pose, text). The student will define metrics for action consistency, completeness, and semantic accuracy, and implement a dashboard that visualizes quantitative assessments alongside interpretable feedback and visual overlays.

[QZ-SP] Student Proposed Project

Detail here if you 're happy to consider student proposed projects and the subject areas you are interested in supervising.

Sean He 's Project Ideas

General Information

I am leading the Computer Vision and Intelligent Perception Laboratory and the Municipal Key Discipline of Computer Science and Technology at UNNC. My current research interests mainly fall into the areas of medical image processing, computer vision, and machine learning.

- Email: sean.he@nottingham.edu.cn
- Tel: ext. 8900

Projects

[SH-1] Automated Autism Spectrum Disorder (ASD) Diagnosis in Children Through Video Analysis

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to develop an innovative, AI-driven system for the early and accurate diagnosis of Autism Spectrum Disorder (ASD) in children by analyzing behavioral cues from video recordings. Traditional ASD diagnostic methods rely heavily on clinical observations and parent-reported questionnaires, which can be time-consuming and subjective. Leveraging advanced computer vision and machine learning techniques, this project will automatically detect and interpret key ASD-related behaviors—such as facial expressions, eye contact, repetitive movements, and social responsiveness—from video data. The system will assist healthcare professionals in making faster, more objective, and data-supported diagnostic decisions, ultimately improving early intervention outcomes for children with ASD. The project objectives include:

1. Developing a Robust Video Analysis Framework. Create algorithms capable of extracting and classifying behavioral markers associated with ASD from video recordings of children in natural or structured settings.
2. Improving Diagnostic Accuracy. Train deep learning models to recognize patterns indicative of ASD with high sensitivity and specificity, reducing reliance on subjective assessments.
3. Enhancing Early Detection. Enable earlier identification of ASD by analyzing subtle behavioral cues that may be missed in conventional screenings.

[SH-2] Generalizable Nuclei Segmentation and Classification in Histopathology

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Histopathological cell segmentation is a cornerstone task in digital pathology analysis, aiming at locating and segmenting individual cell nuclei and determining their cell types (e.g. tumour cells, lymphocytes) from Whole Slide Imaging (WSI). Accurate nucleus classification and segmentation can help to reveal the spatial heterogeneity characteristics of the tumour microenvironment (e.g., degree of lymphocyte infiltration, stromal cell distribution pattern), and provide a quantitative basis for molecular typing of cancers, prognostic assessment, and prediction of therapeutic response. In recent years, deep learning-based histopathological cell intelligent segmentation technology has been gradually applied to this field. This project aims to design a novel nuclei segmentation foundation model that can segment any nuclei instance and type by leveraging flexible Interactive prompts.

[SH-3] An Auxiliary Diagnosis System for Depression Based on Affective Computing

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The purpose of this study is to establish an affective computing-assisted depression diagnosis system. The system will use computer vision, machine learning, and deep learning technologies to automatically identify multimodal data (including physiological data and behavioral data) of depression patients and healthy controls in emotion induction experiments, and quantitatively analyze the differences in emotional responses between depression patients and healthy controls when facing specific emotions, thereby assisting in the diagnosis of depression. The system will use human-computer interaction technology to provide patients with innovative diagnostic methods, and provide a more convenient and comfortable diagnostic experience. The project encompasses several stages, including data acquisition, data processing, model development, data analysis, system development and design, etc.

[SH-4] Multi-organ Segmentation Based on 3D Medical Imaging

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Large pre-trained models for medical image segmentation have garnered significant attention in recent years, exemplified by the release of Med-SAM in 2024. However, with the continued advancement of deep learning techniques, current research has increasingly focused on 3D and temporal data for medical image segmentation. In this study, we aim to design and develop a state-of-the-art model tailored for multi-organ 3D image segmentation. Building upon the theoretical foundation of SAM2, we further explore the integration of multimodal data as prompts to facilitate more effective and contextually appropriate applications.

[SH-5] Foundation Model-based Multi-modal Survival Prediction

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project is focused on developing or optimizing a deep-learning framework based on foundation models to perform survival prediction using multi-modal medical data. It involves leveraging large pre-trained models (e.g., vision-language models or transformer-based architectures) and integrating diverse data modalities such as pathology images, genomic profiles, and clinical records. The project includes various stages: data pre-processing, model adaptation or fine-tuning, survival metric computation (e.g., concordance index, time-dependent AUC), and in-depth data analysis. The goal is to improve the accuracy and interpretability of survival prediction models and to explore the benefits of foundation models in precision medicine applications.

[SH-6] Ultrasound Image Segmentation and Metrics Calculation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project is tasked with developing or optimizing a deep-learning model to segment lesion areas within ultrasound images. They may utilize datasets such as those pertaining to thyroid nodules, breast cancer, Cesarean Scar Defect, and other conditions. The project encompasses several stages, including data pre-processing, model development, measurement index computation, data analysis, etc. The objective of this project is to enhance the understanding of ultrasound image analysis techniques through practical application and to furnish valuable methodologies and research tools for clinical studies.

[SH-7] Cross-Domain Medical Image Translation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Medical image analysis often faces the challenge of domain shift, where models trained on data from one domain (e.g., hospital, imaging modality, or devices) perform poorly when applied to another. This is further complicated by the scarcity of annotated data in many clinical settings. This project aims to investigate the use of diffusion models for cross-domain medical image translation in few-shot scenarios. It consists of four key tasks: data pre-processing, designing domain adaptation pipelines, conditional generation strategies, and evaluating image quality across domains. The goal is to develop a generalizable framework that enables robust medical image translation across different domains with minimal supervision. This framework could improve the adaptability of clinical AI models and enhance performance in low-resource medical environments.

[SH-8] Medical Image Cross-modality Translation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Medical imaging plays a crucial role in diagnosis and treatment planning, but acquiring high-quality images often involves trade-offs between radiation exposure, scan time, and cost. Therefore, cross-modality medical image translation becomes an effective method to address real-world limitations in medical imaging. The related tasks may include:

1. MRI Sequence Translation. Different MRI sequences require separate scans, increasing acquisition time. Synthesizing missing MRI sequences (e.g., generating T2-weighted images from T1-weighted scans) can reduce scan time and improve workflow efficiency.
2. CT-MRI Bidirectional Synthesis. CT and MRI provide complementary information but are not always available together.
3. Ultrasound-to-MRI Translation. Ultrasound is real-time but lacks soft-tissue contrast compared to MRI.

[SH-9] Deep Learning for Medical Image Denoising

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Medical images such as PET, CT, and MRI are essential in clinical diagnosis but often suffer from noise due to low-dose protocols or hardware limitations. This project focuses on developing deep learning-based denoising techniques—such as diffusion models, Transformers, and convolutional networks—to enhance image quality without increasing patient risk. Students will work with real clinical datasets, learn to build and train models using frameworks like PyTorch, and explore how AI can be applied to improve diagnostic imaging. This project is ideal for students interested in AI, image processing, and interdisciplinary applications in medicine.

[SH-10] Real-time Breast Lesion Segmentation in Handheld Ultrasound Using Lightweight Deep Learning

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on developing a lightweight deep learning model to achieve real-time segmentation of breast lesions in handheld ultrasound devices. Targeting clinical scenarios where rapid diagnosis is critical, the framework will enable portable ultrasound systems to provide immediate lesion localization and morphological analysis. The project has several stages including Data Preparation and Pre-processing, Model Development and Optimization, Quantitative Metric Calculation, Validation and Clinical Integration. The goal of this project is to assist clinicians in diagnosis, surgical planning, and treatment monitoring by providing precise tumor localization and quantitative metrics.

[SH-SP] Student Proposed Project

Detail here if you ’re happy to consider student proposed projects and the subject areas you are interested in supervising.

Tianxiang Cui 's Project Ideas

General Information

My general research interests lie primarily in the following areas:

- Computational Intelligence
- Operation Research
- Machine Learning
- Reinforcement Learning
- Computational Finance

You can contact me via:

- Email: tianxiang.cui@nottingham.edu.cn
- Office: PMB-426

Projects

[TC-1] Reinforcement Learning Based Adaptive Alpha Factor Generation

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project employs reinforcement learning (RL) to automate the discovery of alpha factors—predictive signals for financial returns. Traditional factor design relies on static rules, limiting adaptability to shifting markets. Here, an RL agent learns to generate and optimize factors by interacting with market data, aiming to maximize risk-adjusted returns. The framework dynamically balances profitability, stability, and diversification, back-testing against benchmarks like momentum or value strategies. Challenges include handling market noise and avoiding overfitting. Outcomes highlight RL's ability to create adaptive, data-driven factors, offering a modern alternative to conventional quant methods. Ideal for exploring AI-driven finance innovations.

[TC-2] Harnessing Sentiment Analysis with LLMs for Enhanced Trading Strategies

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on exploring the capabilities of different large language models (LLMs) in extracting sentimental features. Utilizing a dataset from Kaggle (https://www.kaggle.com/datasets/sbhatti/financial-sentiment-analysis), we will evaluate the performance of various LLMs in identifying and analyzing these features. The sentimental insights obtained will then be integrated into a reinforcement learning framework, which will aid in training an effective trading policy. By combining advanced sentiment analysis with reinforcement learning, the project aims to enhance decision-making processes in trading, leveraging emotional and market sentiment to inform strategy development and execution.

[TC-3] Strategic Carbon Trading: Leveraging LLMs and Reinforcement Learning for Policy-Driven Markets

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project seeks to develop a sophisticated strategy for carbon emissions trading, a market distinctively influenced by government policies rather than traditional financial metrics. By leveraging large language models (LLMs), we aim to analyse and extract critical features from governmental policy documents and statements. These insights will be integrated into a reinforcement learning framework, which will be instrumental in training an adaptive and effective trading policy. The project aspires to navigate the complexities of carbon markets by aligning trading strategies with the evolving regulatory landscape, thereby enhancing decision-making and optimizing trading outcomes in response to policy shifts.

[TC-4] Development of a Digital Twin System and RMFS Simulator for Autonomous Mobile Robots Using NVIDIA Isaac Sim

- Suitable for CSAI students: Yes
- Project type: Software Engineering

This undergraduate capstone will design and implement a high-fidelity digital twin framework for Autonomous Mobile Robots (AMRs) integrated with a Robotic Mobile Fulfillment System (RMFS) simulator, leveraging NVIDIA Isaac Sim. The student will:

1. Model AMR kinematics, dynamics, and sensor payloads in Isaac Sim;
2. Build a virtual warehouse environment with shelving units, conveyors, and order-fulfillment workflows;

3. Implement real-time synchronization between physical robot telemetry (or recorded logs) and their virtual counterparts;
4. Develop user interfaces and APIs for scenario configuration, execution control, and performance logging;
5. Validate the system through benchmark RMFS tasks (e.g., pick-and-place throughput, path-planning efficiency).

Key deliverables include the complete digital twin platform, a suite of RMFS simulation scenarios, user documentation, and an evaluation report comparing simulated vs. real-world or recorded performance metrics. This project emphasizes software architecture, ROS/Isaac Sim integration, and systematic testing of AMR behaviors within a warehouse fulfillment context.

[TC-5] Hybrid Reinforcement Learning and Operations-Research Solver Framework for variant VRP

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This undergraduate research project investigates the integration of modern Reinforcement Learning (RL) techniques with classical Operations-Research (OR) solvers to tackle a variant of the Vehicle Routing Problem (VRP). The student will:

1. **Literature & Benchmarking:** Conduct a concise survey of RL-based CO methods (e.g. Deep Q-Networks, Policy Gradient) alongside exact and heuristic OR approaches for the chosen VRP variant.
2. **Framework Design:** Build a modular Python framework that couples an RL agent (using PyTorch or TensorFlow) with solver APIs (e.g. Gurobi, CPLEX, or OR-Tools) to handle dynamic decision stages in routing.
3. **Algorithm Development:**
    - Train RL policies to propose promising routing subtours, then refine routes with MIP solvers.
    - Incorporate variant-specific constraints (e.g., time windows, heterogeneous fleets) into both RL reward structures and solver formulations.
4. **Experimental Evaluation:** Test on benchmark datasets for the selected VRP variant, measuring solution quality (total distance, on-time deliveries), computation time, and adaptability to demand fluctuations.

5. **Analysis & Reporting:** Compare pure RL, pure OR, and hybrid methods; document trade-offs, strengths, and limitations in a concise technical report.

> Skills on PyTorch, OR modeling and solvers (Gurobi, CPLEX, OR-Tools) and basic understanding of VRP problem are preferable.

[TC-6] Reinforcement Learning–Based Controller for a Two-Wheeled Wheel-Legged Robot

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This FYP focuses on designing and evaluating a Reinforcement Learning (RL) controller for a novel wheel-legged robot platform that combines the speed of wheels with the adaptability of legged locomotion. The student will:

1. **Simulation Environment:** Build a realistic robot model in Nvidia ISAAC LAB, including wheel-leg kinematics and dynamics, to serve as a training playground.
2. **RL Algorithm Development:** Implement state-of-the-art continuous-action RL methods (e.g., DDPG, PPO) to learn locomotion policies that seamlessly switch between rolling and stepping modes.
3. **Reward Engineering:** Craft reward functions that balance forward velocity, energy efficiency, and stability during mode transitions and on uneven terrain.
4. **Training & Evaluation:** Train agents in progressively challenging environments (flat ground, slopes, obstacles), then benchmark performance against a baseline PID-based controller in terms of speed, stability (e.g., foot slip, tip-over rate), and energy consumption.
5. **Sim-to-Real Validation:** Deploy the learned policy on a physical prototype or hardware-in-the-loop setup to assess transferability and robustness.

Key challenges include managing high-dimensional continuous control, ensuring smooth hybrid gait transitions, and mitigating the sim-to-real gap for reliable real-world operation.

[TC-7] Spatial-Temporal Electric Vehicle Charging Demand Prediction

- Suitable for CSAI students: Yes
- Project type: Empirical Research

T-EVCDP is a real-world dataset designed for predicting electric vehicle (EV) charging demand, primarily aimed at spatio-temporal data analysis, charging demand forecasting, and urban energy optimization. The dataset is derived from extensive charging station data in Shenzhen, covering:

- 18,061 public charging stations
- 247 traffic zones
- 1,006 neighboring area connections
- 30 days of charging data sampled every 5 minutes

The data is stored in a graph structure, with each charging station, area, and their neighboring relationships modeled to facilitate in-depth analysis by researchers and engineers.

The dataset is designed to support predictions using large language models (LLMs), which are advanced artificial intelligence systems capable of understanding and generating human-like text. By leveraging natural language processing (NLP) methods, the project aims to explore innovative ways to utilize LLMs for predicting EV charging demand. This involves analyzing complex patterns and trends in the dataset and using LLMs to interpret and forecast future charging needs. As part of the ChatEV research initiative, this approach aims to harness the capabilities of LLMs to improve accuracy and efficiency in demand forecasting, potentially transforming how urban planners and energy providers manage and optimize electric vehicle charging infrastructure. Through this integration, the dataset not only serves traditional analytical purposes but also opens new avenues for AI-driven insights and solutions in sustainable urban mobility.

[TC-SP] Student Proposed Project

I am also happy to consider student proposed projects in the areas of Reinforcement Learning Computational Finance and Operation Research.

Wooi Ping Cheah 's Project Ideas

General Information

The main focus of my research is currently in knowledge engineering, causal reasoning, machine learning, and artificial intelligence applications.

- Email: Wooi-Ping.Cheah@nottingham.edu.cn
- Office: PMB323

Projects

[WP-1] An Intelligent Advising System for University Majors with Explanation Capabilities

- Suitable for CSAI students: Yes
- Project type: Software Engineering

This project aims to develop an intelligent advising system to assist university applicants in selecting majors aligned with their academic strengths, interests, and career goals. The system will incorporate AI-driven recommendations and an explanation module to justify its suggestions, enhancing transparency and trust. The following tasks are to be done in this project:

- Data Collection: Gather student profiles (grades, interests, skills) and labour market trends.
- Machine Learning Model: Train a recommendation algorithm to suggest suitable majors.
- Explanation Facility: Implement natural language generation to clarify recommendations (e.g., ''Your analytical skills match Computer Science'').
- User Interface: Design a web/mobile platform for interactive exploration.

This system bridges the gap between data-driven guidance and human-centric counselling, helping students to make informed decisions. An expected outcome is a personalised, explainable advising tool, which will improve major satisfaction and higher retention rates by matching students with optimal academic paths.

[WP-2] Financial Fraud Analysis Using Bayesian Networks

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Financial fraud poses significant risks to businesses and individuals, demanding advanced detection methods. This project proposes a Bayesian network-based approach to analyse and identify fraudulent transactions by modelling probabilistic relationships between variables. This project bridges AI and financial security, offering a scalable, interpretable tool for fraud analysts. The specific objectives of this project are:

- Develop a probabilistic model using Bayesian networks to detect anomalies in financial data.
- Evaluate the model's accuracy against traditional methods.
- Provide interpretable results through conditional probability analysis to aid fraud investigators.

The following tasks are to be done in this project:

- Data Collection: Use labelled datasets (e.g., Kaggle's credit card fraud data).
- Structure Learning: Identify the probabilistic dependencies between variables in a Bayesian network by analysing data to infer the most likely graphical model.
- Parameter Learning: Estimate the conditional probability distributions for each node in the network, optimizing them to fit observed data while quantifying uncertainty.
- Validation: Test performance metrics (precision, recall, F1-score) against real-world fraud cases.

An expected outcome is a transparent, explainable fraud detection system using Bayesian probability.

[WP-3] Enhancing E-Commerce Recommendations with Large Language Models

- Suitable for CSAI students: Yes
- Project type: Software Engineering

This project aims to improve e-commerce recommendation systems by leveraging large language models (LLMs) to generate personalised and context-aware product suggestions. Traditional recommendation systems rely on collaborative filtering or content-based methods, which may lack semantic understanding. LLMs can analyse user queries, reviews, and browsing behaviour to provide more accurate and explainable recommendations. The following tasks are to be done in this project:

- Data Collection: Gather user interactions, product descriptions, and reviews from e-commerce platforms.
- Model Fine-Tuning: Adapt a pre-trained LLM (e.g., GPT, DeepSeek) using e-commerce-specific datasets.

- Recommendation Generation: Implement a hybrid approach combining LLM-based semantic analysis with traditional collaborative filtering and content-based methods.
- Evaluation: Compare the performance of the LLM-enhanced system against baseline models using metrics like precision, recall, and user satisfaction surveys.

The expected outcome of this project is a recommendation system that improves user engagement, reduces irrelevant suggestions, and enhances interpretability through natural language explanations.

[WP-4] Email Spam Filtering Using Large Language Models

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project aims to develop a spam filtering system using pre-trained Large Language Models (LLMs) to improve detection accuracy and adaptability compared to traditional methods (e.g., Naïve Bayes, SVM). The focus will be on exploiting LLMs' semantic understanding to identify sophisticated spam patterns, including phishing and adversarial evasion techniques. The following tasks are to be done in this project:

- Dataset: Use benchmark datasets and augment with real-world emails.
- Model Selection: Experiment with pre-trained LLMs (e.g., BERT, GPT-3.5/4, or open-source alternatives like DeepSeek).
- Evaluation: Compare performance (precision/recall/F1-score) against classical ML approaches.

The outcomes of this research aim to contribute to the development of more effective email spam filters, reducing the impact of unwanted emails on users and organisations.

[WP-SP] Student Proposed Project

Students are welcome to propose and discuss their project ideas with me, covering knowledge engineering, machine learning, programming paradigms and other related research areas.

Xinan Chen 's Project Ideas

General Information

My research interests encompass Operations Research, Reinforcement Learning, Evolutionary Algorithms, Large Language Models, and Data Science, with a specialization in applying advanced artificial intelligence techniques to solve complex, real-world optimization problems. Students who select my FYP topics will have the opportunity to join my research team and collaborate with my PhD and Master students. Exceptional student work may receive support for publication in conference proceedings or journals. Prior to joining the research team, students should possess fundamental knowledge of machine learning and operations research.

You can contact me via:

- Email: xinan.chen@nottingham.edu.cn
- Office: PMB 442

Projects

[XC-1] Large Language Model Support Decision Making System in Jobshop Scheduling

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project explores how Large Language Models (LLMs) can be integrated into or used to support the decision-making process in complex job shop scheduling environments. The goal is to leverage LLM capabilities, such as interpreting natural language constraints, explaining complex schedules, or assisting human schedulers, to potentially make the challenging task of job shop scheduling more intuitive, efficient, or understandable.

[XC-2] Static Scheduling in Container Port Truck Dispatching

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project involves modeling and solving the problem of scheduling internal trucks within a container terminal for a fixed set of known container movements. The goal is to use static scheduling methods like Genetic Algorithms (GA), Ant Colony Optimization (ACO), and Simulated Annealing (SA) to optimize truck assignments and routes, typically aiming to minimize total travel time, waiting time, or maximize throughput under predefined conditions over a specific time horizon.

[XC-3] Dynamic Two Yard Crane Scheduling in Container Port

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project specifically addresses the dynamic scheduling challenge for two yard cranes operating collaboratively within the same yard block of a container terminal. The core task is to use AI methods to develop a system that can make real-time decisions on job sequencing and movement coordination for the two cranes, ensuring efficient container handling while actively avoiding collisions and deadlocks in their shared operational space.

[XC-4] Optimization of Optical Device Design

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project focuses on applying computational optimization and potentially AI methods to automate or enhance the design process of optical devices. By integrating with optical simulation tools, the aim is to use algorithms to efficiently search the vast parameter space of device designs and find optimal configurations that achieve desired performance characteristics, thereby accelerating and improving the quality of the design process.

Project: Student Proposed Projects

[XC-SP] Student Proposed Project

I welcome students to propose their own Final Year Project ideas! This track encourages students to explore topics within Computer Science that align with their personal interests and career aspirations. Proposed projects should clearly define a problem, outline a feasible approach, specify expected deliverables (such as a software system, analysis, or research findings), and demonstrate the student's motivation and capability to undertake the work within the project timeframe.

Yuan Cheng 's Project Ideas

General Information

Security and privacy, with an emphasis on access control and authentication.

Please contact me by email

- Email: yuan.cheng@nottingham.edu.cn
- Office: PMB442

Projects

[YC-1] Fast and Furious, but at What Cost? Evaluating the Performance and Privacy Implications of HTTP/3 over QUIC

- Suitable for CSAI students: No
- Project type: Empirical Research

HTTP/3, powered by the QUIC protocol, promises faster web browsing speed and improved responsiveness compared to its predecessors, HTTP/2 and HTTP/1.1. However, its adoption also brings new challenges in security and privacy that remain largely unexplored. In this project, students will conduct an empirical analysis of both the performance gains (e.g., latency, throughput, page load times under various network conditions) and the privacy implications (e.g., traffic fingerprinting, data leakage, and potential tracking risks) associated with deploying HTTP/3 over QUIC. The goal of the project is to enable students to critically assess the trade-offs between cutting-edge network protocols and user privacy.

[YC-2] Not So Private After All? A Forensic Investigation of Private Browsing Modes across Web Browsers and Platforms

- Suitable for CSAI students: No
- Project type: Empirical Research

Private browsing modes in popular web browsers (Edge, Chrome, Firefox, Safari) promise users a private and confidential web experience by claiming to leave no trace of browsing activity. However, do these modes truly erase all evidence of user activity, or do they leave behind inadvertent traces? In this empirical research project, students will conduct a forensic investigation across multiple browsers and platforms (PC, Mac, iOS, Android) to identify potential privacy leaks in memory dumps, file systems, browser caches, cookies, network logs, and DNS queries. By systematically examining these channels, the project aims to evaluate the privacy claims of browser vendors, identify discrepancies, and propose actionable recommendations for safer browsing practices.

[YC-3] MingLi (名理)Intelligent Cross-Cultural Name Generator Powered by Large Language Models

- Suitable for CSAI students: Yes
- Project type: Software Engineering

Choosing a name for a child across different cultures can be both exciting and challenging. Parents often seek names that beautifully blend pronunciation, meaning, and cultural significance. This project focuses on building a web service/app named *MingLi* (名理 ,名 = Name ,理 = Reason), which uses state-of-the-art large language models (LLMs, such as GPT and DeepSeek) to intelligently suggest name conversions between English (or other languages beyond the Sinosphere) and Chinese. Students will incorporate AI-driven natural language processing techniques, semantic reasoning, and an understanding of cultural context into the service, enabling it to generate culturally nuanced and linguistically elegant name suggestions. The resulting service will serve as an invaluable resource for expatriates, multicultural families, students, and professionals navigating cross-cultural interactions.

[YC-4] AppPulse: Sentiment Analysis and Feature Insights from App Store Reviews

- Suitable for CSAI students: Yes
- Project type: Software Engineering

Understanding user sentiment and identifying emerging trends from app store reviews are essential for developers seeking to improve their products and prioritize updates. In this project, students will create an intelligent sentiment analysis and visualization tool for app store reviews from Apple App Store and Google Play. The system will enable users to filter and analyse feedback based on various criteria, including app name, language, version, date, and rating. By employing natural language processing techniques, the tool will identify the most requested features, common complaints, and urgent bug reports. An interactive dashboard will visualize sentiment trends over time, keyword frequencies, and heatmaps of user concerns, allowing engineering teams to make informed decisions quickly and effectively.

[YC-5] Building and Comparing Real-World and Synthetic Passphrase Datasets

- Suitable for CSAI students: Yes

- Project type: Empirical Research

Passphrases are often regarded as a more user-friendly and secure alternative to passwords. However, the lack of publicly available passphrase datasets limits further research and development in this field. This project asks students to create a curated dataset of passphrases using two complementary methods. First, students will analyse leaked password datasets to extract user-chosen long passwords, then systematically study them for linguistic and structural patterns. Second, students will employ Markov models and/or large language models (LLMs) to synthetically generate passphrases that resemble natural language. The resulting datasets will then be compared based on factors such as entropy, syntax, semantics, and memorability. This dual-dataset approach will provide valuable insights into how real-world passphrases differ from those generated by machines, supporting future research aimed at developing more secure and memorable authentication methods.

[YC-6] A Mobile Phishing Awareness App Powered by AI for Senior Citizens

- Suitable for CSAI students: Yes
- Project type: Software Engineering

Phishing remains one of the most prevalent cybersecurity threats, particularly for elderly users who may lack digital literacy and awareness of evolving scam tactics. This project aims to design and build a mobile app (and/or a WeChat mini-program) that delivers personalized, interactive phishing awareness training tailored for senior citizens. Powered by large language models (LLMs), the app will simulate realistic phishing scenarios via interactive messages, fake login pages, and mimicked SMS, emails, and bot calls, all within a safe, gamified environment. To enhance the learning experience, the system will feature AI-generated educational videos with conversational narrators and storytelling elements to explain common phishing tricks and red flags. These short videos will be incorporated into a progressive training curriculum that includes quizzes, user-friendly feedback, and adaptive difficulty levels. The goal of the app is to empower users with essential cybersecurity skills through an engaging and effective learning experience.

[YC-SP] Student Proposed Project

I welcome student proposed project ideas. Before reaching out, please take the time to thoroughly explore my homepage and read at least one recent research paper. If doing this doesn't give you any interesting ideas, it may be best to seek guidance elsewhere. However, if this does, please send a short introductory email that includes your resume and a concise 1-page project abstract.

Ying Weng 's Project Ideas

General Information

My Research Interests –Machine Learning, Mixed Reality, Computer Vision, Image/Video Processing.

If you are interested in any of the following topics, or any topics related to them, please make an appointment with me by email for a discussion.

General requirements:

- Continuous and consistent dedication to the project: You will be expected to be self-motivated, hardworking, and allocate a reasonable time for the project every week, from the beginning to the end of the academic year.
- Weekly meetings: You will be expected to have weekly meetings with the supervisor. The result-target format will be used. Your previous result will be discussed and your next target will be set up during each meeting.

The more specific requirements for each project shall be discussed personally with each applicant.

- Email: Ying.Weng@nottingham.edu.cn
- Office: PMB341

Projects

[YW-1] Multi-source Data Fusion based Intelligent Water Ecology Management

- Suitable for CSAI students: Yes
- Project type: Empirical Research

The concept of sustainable cities has gained significant attention with the rapid growth of urban areas and the increasing challenges posed by environmental change and resource scarcity. Sustainable cities are those that prioritize the efficient use of resources, minimize environmental impact, and enhance the quality of life for their residents. To effectively address these challenges and achieve sustainable urban development, it is essential to harness the power of multi-source data fusion techniques. Multi-source data fusion plays a crucial role in sustainable cities by providing a comprehensive and holistic understanding of urban environments. However, existing approaches often fall short in building sustainable cities due to their limitations in their scope and coverage. Furthermore, sustainable urban development requires a multidisciplinary approach, considering social, economic, and environmental factors. To overcome these limitations and advance sustainable urban development, multi-source data fusion techniques provide a promising solution. Intelligent water ecology management is of great importance to water ecological construction, which allows to exhibit temporal and spatial dynamic behaviour. By integrating data from various sources, we can gain valuable insights into urban systems and implement effective strategies for sustainable urban planning and management.

[YW-2] Empathic Interaction Design in VR System for ADHD Children

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder that affects a significant number of children, impacting their attention, impulse control, and hyperactivity. Virtual Reality (VR) technology has emerged as a promising tool for intervention, and empathic interaction design in VR systems can enhance its effectiveness for ADHD children.

Empathic interaction design is crucial as it allows the VR system to connect with the child on an emotional level. The avatars and characters within the VR environment can be designed to display empathetic behaviors. They can offer words of comfort, celebrate small achievements, and show understanding when the child makes mistakes. This empathetic connection helps build trust between the child and the VR system, making the child more engaged and receptive to the therapeutic or educational content.

[YW-3] AI-Driven Analysis and Optimization of Performance on Magnets

- Suitable for CSAI students: Yes
- Project type: Empirical Research

This project will carry out collaborations with Magnetic Material Company. The project objectives include: (i) Developing an AI-empowered automatic identification system for magnet nanograin characterization; (ii) Constructing predictive models for grain boundary diffusion depth via machine learning algorithms; (iii) Mapping process-microstructure-property correlations to guide magnetic performance optimization. The project implementation offers hands-on experience in AI-driven materials informatics, computational microstructure analysis, and data-intensive magnet design.

[YW-4] Can DeepSeek Provide Useful Code Feedback?

- Suitable for CSAI students: Yes

- Project type: Empirical Research

DeepSeek interacts in a conversational way. The dialogue format makes it possible for DeepSeek to answer follow-up questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. DeepSeek can produce both useful and unusable code. For best results, provide clear and detailed prompts. DeepSeek excels in assisting with specific coding tasks or routines, rather than building complete applications from scratch. When DeepSeek generates code suggestions, take the opportunity to study and understand the logic behind them. This project will investigate: Can DeepSeek provide useful code feedback through the real scenarios?

[YW-SP] Student Proposed Project

I am happy to consider student proposed projects and the subject areas I am interested in supervising. Please email me to make an appointment for a discussion.

Yuan Yao 's Project Ideas

General Information

My research interests are in Artificial Intelligence, in particular the area of Autonomous Agents and Multi-agent Systems including the following topics: Agent Reasoning (How can agents behave in smart ways), Agent Learning (How can agents learn from the environment), Agent Architecture and Agent-Oriented Programming Languages (How to effectively develop agent-based systems), Logics and Verification (How to verify and validate large-scale Multi-agent Systems)

Students can contact me via email or come to my office during my office hours.

- Email: Yuan.Yao@nottingham.edu.cn
- Office: PMB438

Projects

[YY-1] Goal Recognition in Multiagent Systems

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Goal recognition involves identifying the intentions of other agents by inferring them based on observed actions or effects within the environment. Goal recognition has been shown to be useful in a wide range of scenarios, including story understanding, human-computer interaction, traffic monitoring, and so on. This project focuses on addressing contemporary challenges in goal recognition, such as the identification of maintenance goals, and aims to design and implement goal recognition algorithms to tackle these issues.

Required skill set: Python, Java, Reinforcement Learning, Probability Theory.

[YY-2] Incorporating LLM into the Belief-Desire-Intention Model

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Recent advances in large language models (LLMs) have significantly broadened their range of applications, surpassing their role as conversational agents, enabling their integration into interactive systems, and accelerating advancements toward artificial general intelligence. The Belief-Desire-Intention-based (BDI) agent model, on the other hand, is one of the most commonly used agent architectures for designing and implementing autonomous agents. In the BDI model, the behaviour of an agent is specified in terms of its mental attitude, i.e., beliefs, desires(goals), and intentions. This project explores the feasibility of integrating these two approaches, either by leveraging the BDI model to enhance the performance of LLMs or, conversely, by utilizing LLMs to improve the effectiveness of BDI agents.

Required skill set: Python, Java, LLM, Agent, BDI agent.

[YY-3] Agent-based Simulations

- Suitable for CSAI students: Yes
- Project type: Software Engineering

Modelling and simulation have long been dominated by equation-based approaches, until the recent advent of agent-based approaches. In this project, the student is asked to develop a software that is capable of simulating agents' behaviour (including goal achieving, cooperation and competition) in a particular real-world scenario. The student can either develop its application from scratch or use existing open-source simulators, e.g., Gazebo (https://gazebosim.org), Carla (https://carla.org/) and so on.

[YY-4] AI for Some Game

- Suitable for CSAI students: Yes
- Project type: Empirical Research

Designing AI players for computer games is one of the most popular ways to validate the effectiveness and efficiency of state-of-the-art AI algorithms. There is a list of very famous AI game competitions available online, including Kaggle, Coder One, MineRL and so on. In this project, the students are asked to create their own AI players for some games and ideally to submit their work to the corresponding AI competition.

Zheng Lu 's Project Ideas

General Information

My projects are mainly focused on machine learning, natural language processing, and computer vision. Once you select one of my projects, you will join my research group and work closely with my PhD students on the related projects. In this way, you can quickly get started with the project and is able to seek help conveniently if needed. We will have weekly group meeting through out the whole year and probably personal meeting every one or two weeks depending on your progress. During group meeting, students are expected to read relevant research papers and share their ideas with each other. During personal meeting, I will discuss with your latest progress and plans for the next week based on current results. For all my projects, students will learn research and programming skills in the AI area which hopefully can help with their future career. You are welcome to drop me an email or make an appointment to discuss the details of my projects.

Please contact me using the following methods for further discussion

- Email: zheng.lu@nottingham.edu.cn
- Office: PMB426

Projects

[ZL-1] Fine-grained Image Caption Generation Using Deep Learning

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

In Computer Vision, image captioning aims to generate text description from a given image, making use of both Computer Vision and Natural Language Processing techniques. Such image description contains usually a few sentences capturing the main content from the image. Fine-grained image caption generation not only generates image descriptions but also capturing fine details that generic image caption techniques does not capture. Such details may be crucial to differentiate images that are similar but differ at subtle details. In this project, You will first learn the state-of-the-art image captioning techniques in addition to general deep learning. Then, you will develop your own algorithm hopefully obtain better image captioning results.

[ZL-2] Object Detection from UAV images

- Suitable for CSAI students: Yes

- Project type: Theoretical Research

The purpose of this project is to develop an object detection from UAV images with novel deep learning based Computer Vision techniques. In Computer Vision, object detection is referred to as a group of algorithms to detect where objects are located in an image. Such techniques are very useful in human daily life such as detecting cars, buses, or human without significant effort of human labelling (human tells the model where the objects are located in an image). UAVs are very popular these days and being able to automatically detect objects from their cameras are crucial to their intelligent functions. In this project, you will first learn the state-of-the-art object detection techniques in addition to general deep learning. Then, you will develop your own algorithm hopefully obtain better object detection results from UAV images.

[ZL-3] Image Enhancement Using Deep Learning Models

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Image enhancement concerns how to automatically enhance a given text input image using Computer Vision techniques. Image enhancement is a very hot topic in the area of computational photography even before the deep learning era. Apps such as Photoshop contains a lot of image enhancement techniques many of which are based on machine learning models. In this project, you are expected to exploit the latest and develop new deep learning based image enhancement techniques. This topic is very open and the student is expect to narrow down to certain problem. For example, how to automatically remove extra objects such as pedestrian for selfie images. Then the student is expected to solve such problem using an existing state-of-the-art method or a new framework.

[ZL-4] Image Generation from Text Description

- Suitable for CSAI students: Yes
- Project type: Theoretical Research

Image generation is defined as generate an image from a given text description. Image generation is a very hot Computer Vision topic especially in current deep-learning era including the latest development of diffusion model. The problem is very challenging as the algorithm not only should be able to generate a sensible image but also need to match the text description as much as possible. For example, if the description is ‘‘Generate an image with mathematical formula and the sea’’, the output image should contain the sea as the main content as well as correct mathematical formula somewhere. In this project, you are expected to exploit the latest and develop new deep learning based computer vision techniques to solve the problem of image generation. There are many existing techniques available for the type of problem that can be used such as diffusion model and its variations. The student needs to implement an existing state-of-the-art method, or build a new framework. Then, extensive experiments on large datasets need to be conducted in order to evaluate the performance gain of the proposed method. The student could also restrict the problem domain to a smaller scope such as artistic images only instead of generic images.

[ZL-SP] Student Proposed Project

I am happy to consider student project proposals in the following areas:

- Natural Language Processing
- Computer Vision
- Image or Video Processing
- Machine Learning

I expect the students approach me through emails and making appointments to discuss about their ideas. However the students should do some background reading before the meeting so that its easier to draw an conclusion whether the topic is good enough for both of us.